2026 prelim electricity questions

Electricity and magnetism questions from the 2026 prelim directory.
Whole pages are kept. A page that also holds another question is included in full.
RI items are the marked scripts. There is no blank RI question paper in the corpus.

2026 RI P2 Q4 p.2
2026 RI P2 Q5 p.5
2026 RI P2 Q8 p.7
2026 RI P3 Q5 p.14
2026 RI P3 Q6 p.16
2026 RI P3 Q7 p.18
2026 RI P3 Q8 p.20
2026 DHS P2 Q4 p.29
2026 DHS P2 Q5 p.30
2026 DHS P2 Q7 p.33
2026 DHS P3 Q1 p.39
2026 DHS P3 Q5 p.41
2026 DHS P3 Q6 p.43
2026 EJC P2 Q3 p.45
2026 EJC P2 Q4 p.47
2026 EJC P2 Q6 p.49
2026 EJC P2 Q8 p.52
2026 EJC P3 Q8 p.57
2026 HCI P2 Q5 p.62
2026 HCI P2 Q6 p.64
2026 HCI P3 Q5 p.67
2026 HCI P3 Q6 p.70
2026 NYJC P3 Q4 p.72
2026 NYJC P3 Q5 p.74
2026 NYJC P3 Q9 p.76
2026 VJC P2 Q5 p.81
2026 VJC P3 Q4 p.84
2026 VJC P3 Q5 p.85
2026 VJC P3 Q9 p.89

4 **(a)** An experiment is performed using two large vertical plates maintained at potential + 6.0 V and – 6.0 V and separated by distance *d* as shown in Fig. 4.1.

When a particle of mass 0.25 g and charge *Q* is released from rest between the plates, it moves in a straight line at an angle of 45° to the vertical.

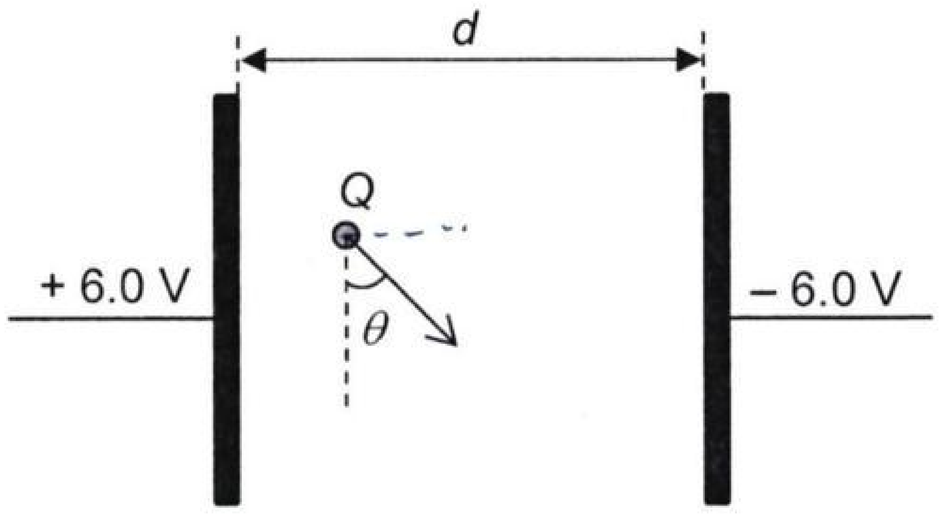


Fig. 4.1

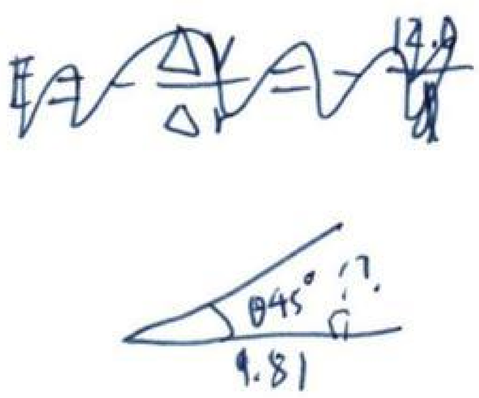
(i) State and explain the sign of the charge on the particle.

1 / 1

Q is positive because it is attracted to negative potential plate, and unlike poles attract. [1] ✓ (1)

(ii) Determine the magnitude of the electric force acting on the charged particle.

0 / 1



by vector resolution for weight & e-force:

electric force = $9.81 \tan 45^\circ$
$= 9.81$

Huh?

electric force = 9.81 N [1]

(b) The distance d between the plates is changed and the experiment is repeated with the particle being released from rest at the same initial position.

The graph of $\tan\theta$ against $\frac{1}{d}$ is shown in Fig. 4.2.

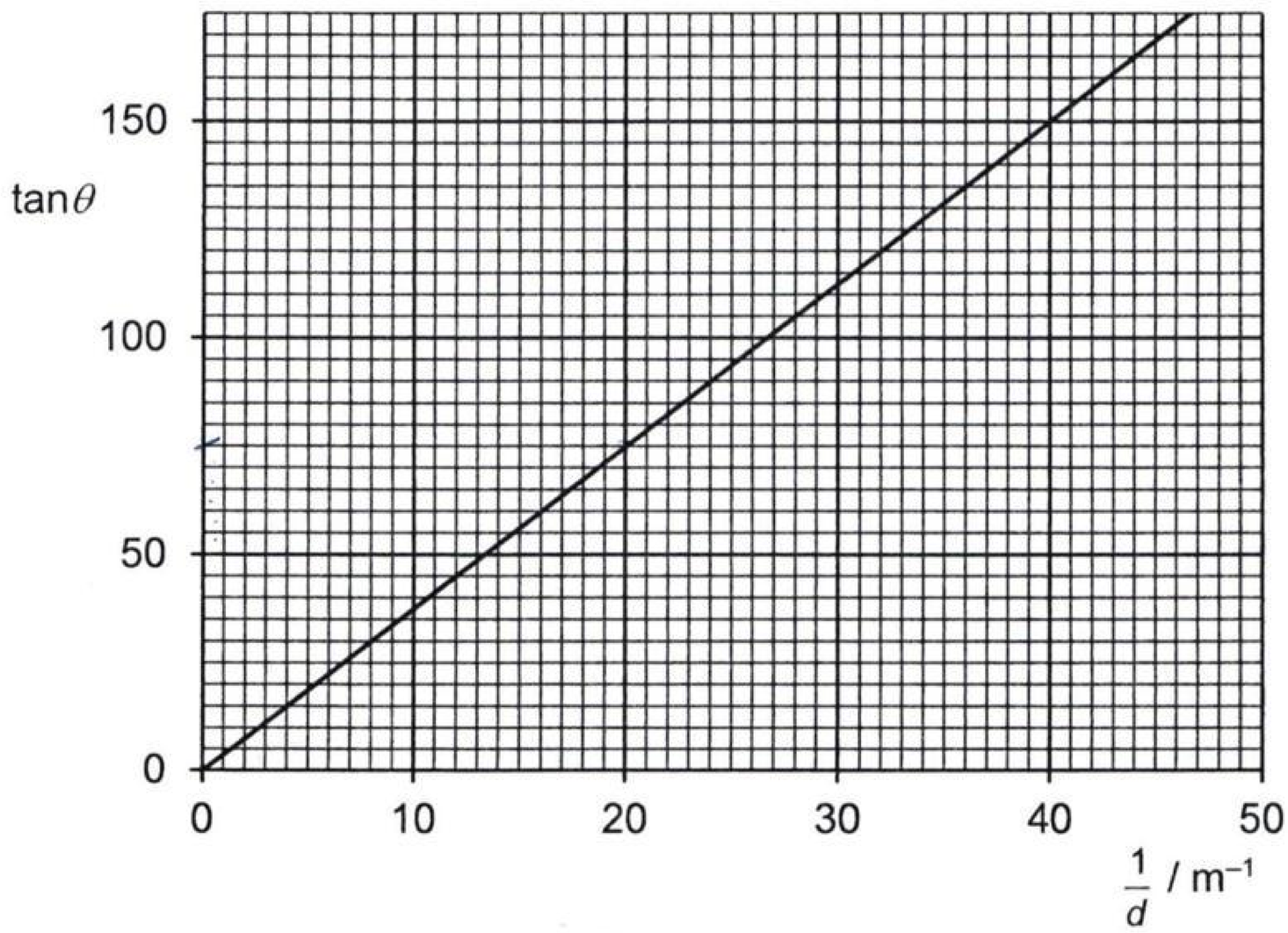


Fig. 4.2

Determine the magnitude of charge Q.

0 / 3

$E = -\frac{\Delta V}{\Delta r} = -\frac{12.0}{d} = (-12.0)\left(\frac{1}{d}\right)$

tan θ is coefficient of e-force ? Huh ?

e-force = QE

at $\frac{1}{d} = 20$, tan θ = 7.5

e- force = (9.81)(75) ?

E = (-12.0)(20)

$Q = \left|\frac{(9.81)(75)}{(-12.0)(20)}\right|$ Q is positive

= 3.0638

= 3.1 C (2 sf)

1 dp for 1/2 small sq

Q = ... 3.1 ..... C [3]

(c) The two large vertical plates in **(a)** form a parallel plate capacitor of capacitance 5.0 pF that is charged to 12.0 V.

This capacitor is then connected to another capacitor with capacitance of 10.0 pF that is initially charged to 6.0 V as shown in Fig. 4.3.

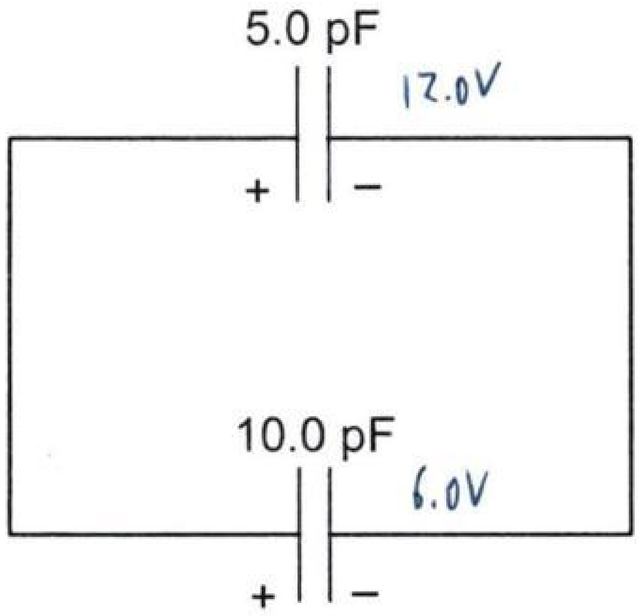


Fig. 4.3

Determine the final charge on each capacitor.

0 / 3

$C = \frac{Q}{V}$

$5.0\ pF = 5.0 \times 10^{-12}\ F$

$10.0\ pF = 10.0 \times 10^{-12}\ F$

Caps in series: $\frac{1}{C_T} = \frac{1}{C_5} + \frac{1}{C_{10}}$

$C_T = \left(\frac{1}{5\times10^{-12}} + \frac{1}{10\times10^{-12}}\right)^{-1}$

$= \frac{10}{3} \times 10^{-12}\ F$

$Q = CV$

$= \left(\frac{10}{3} \times 10^{-12}\right)(12.0)$

$= 4.0 \times 10^{-11}\ C$

charges:

$Q_5 = (5.0 \times 10^{-12})(12.0)$

$Q_{10} = (10.0 \times 10^{-12})(6.0)$

same charge $Q_5 = Q_{10}$

Final charges

$Q = (5.0 \times 10^{-12})(?)$

$Q_{5\,tot} = \frac{1}{2} Q_{5t}$

$= 4 \times 10^{-11}\ C$

$Q_{10\,tot} = \frac{2}{3}(5.0 \times 10^{-12})(12.0)(2)$

$= 8.0 \times 10^{-11}\ C$

final?

charge on the 5.0 pF capacitor = 4.0×10^{-11} C

charge on the 10.0 pF capacitor = 8.0×10^{-11} C [3]

[Total: 8]

5 A length of conducting wire is bent into the shape of an isosceles right-angled triangle, where the two equal sides are each 0.15 m in length as shown in Fig. 5.1.

The loop is placed in a uniform magnetic field of flux density 0.40 T directed perpendicularly into the plane of the loop. The loop moves at a constant velocity of 0.30 m s^{-1} in the plane of the loop to the right.

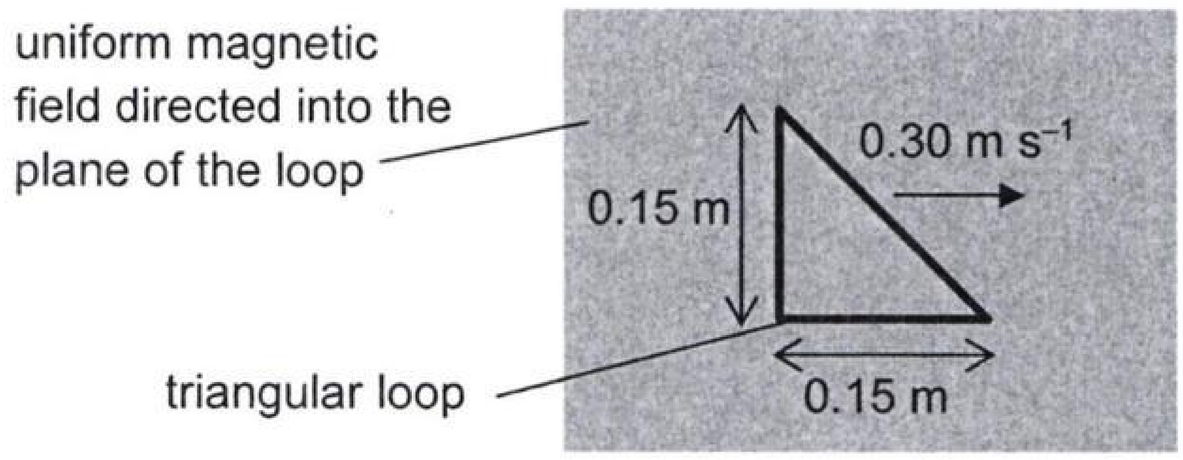


Fig. 5.1

(a) Explain why an e.m.f. is induced in the loop as it moves out of the magnetic field.

0 / 2

By Faraday's law of electromagnetic induction, there is induced e.m.f when there is a change in magnetic flux. When an area of wired loop moves, there is a change in the magnetic flux By the loop and this induces e.m.f.

vague

[2]

(b) The loop starts to exit the field at time $t = 0$ and it fully exits the field at time $t = T$.

On Fig. 5.2, sketch the variation of induced e.m.f. E with time t.

0 / 1

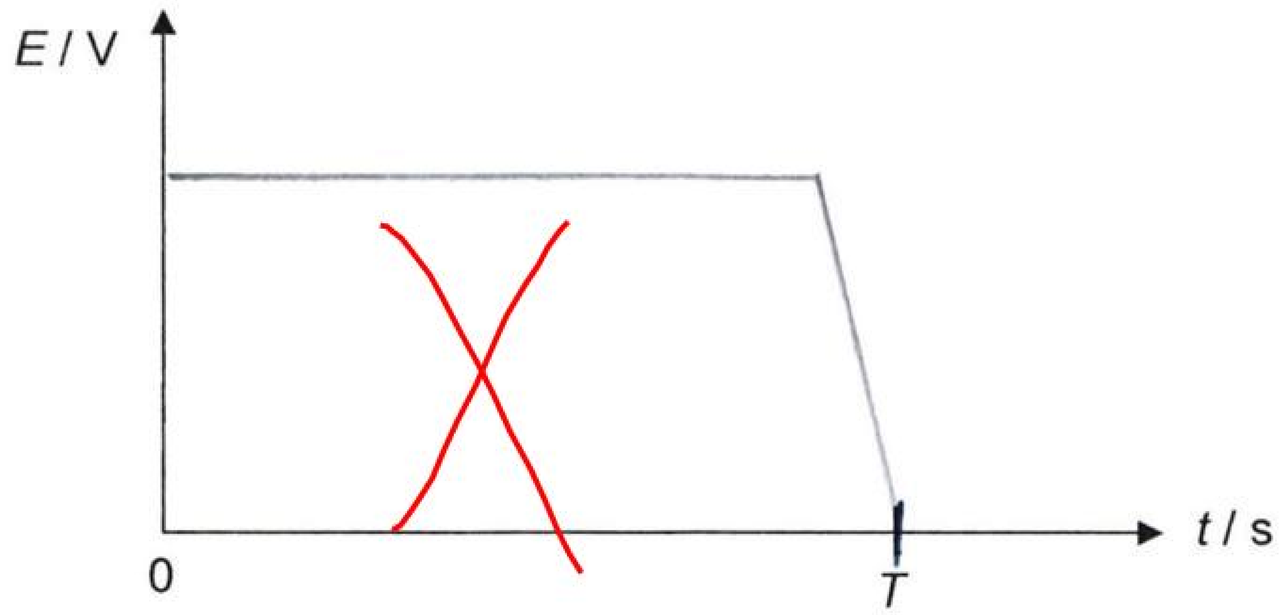


Fig. 5.2

[1]

(c) Determine the average e.m.f. induced in the loop.

0 / 2

e.m.f. = ... V [2]

[Total: 5]

6 **(a)** Describe and explain an observation from the photoelectric effect that provides evidence for the particulate nature of electromagnetic radiation.

0 / 2

.....

.....

.....

.....

.....

.....

..... [2]

(b) Explain how emission line spectra provide evidence for discrete electron energy levels in isolated atoms.

0 / 2

.....

.....

.....

.....

.....

..... [2]

[Total: 4]

8 Read the passage below and answer the questions that follow.

In recent years, the global shift towards electric vehicles (EVs) has been driven by the need for sustainable transportation. EVs are often considered 'greener' and more efficient than conventional internal combustion engine (ICE) vehicles. In terms of energy consumption, a typical EV consumes approximately 18 kW h per 100 km, while a conventional petrol vehicle with an ICE consumes about 7.0 litres of fuel per 100 km. Petrol used in ICE vehicles has an energy density of approximately 34.2 MJ per litre.

An EV has a battery that supplies direct current, which an inverter converts into alternating current to drive its electric motor. In the motor, coils of wire are arranged around a rotating part called a rotor and placed in a magnetic field. When current flows through these coils, magnetic forces act on opposite sides of the coil in opposite directions, producing a torque to spin the motor. Fig. 8.1 shows one of the coils.

As the motor spins, a back electromotive force (e.m.f.) is induced in the coils which oppose the supply voltage and plays an important role in regulating the current drawn by the motor.

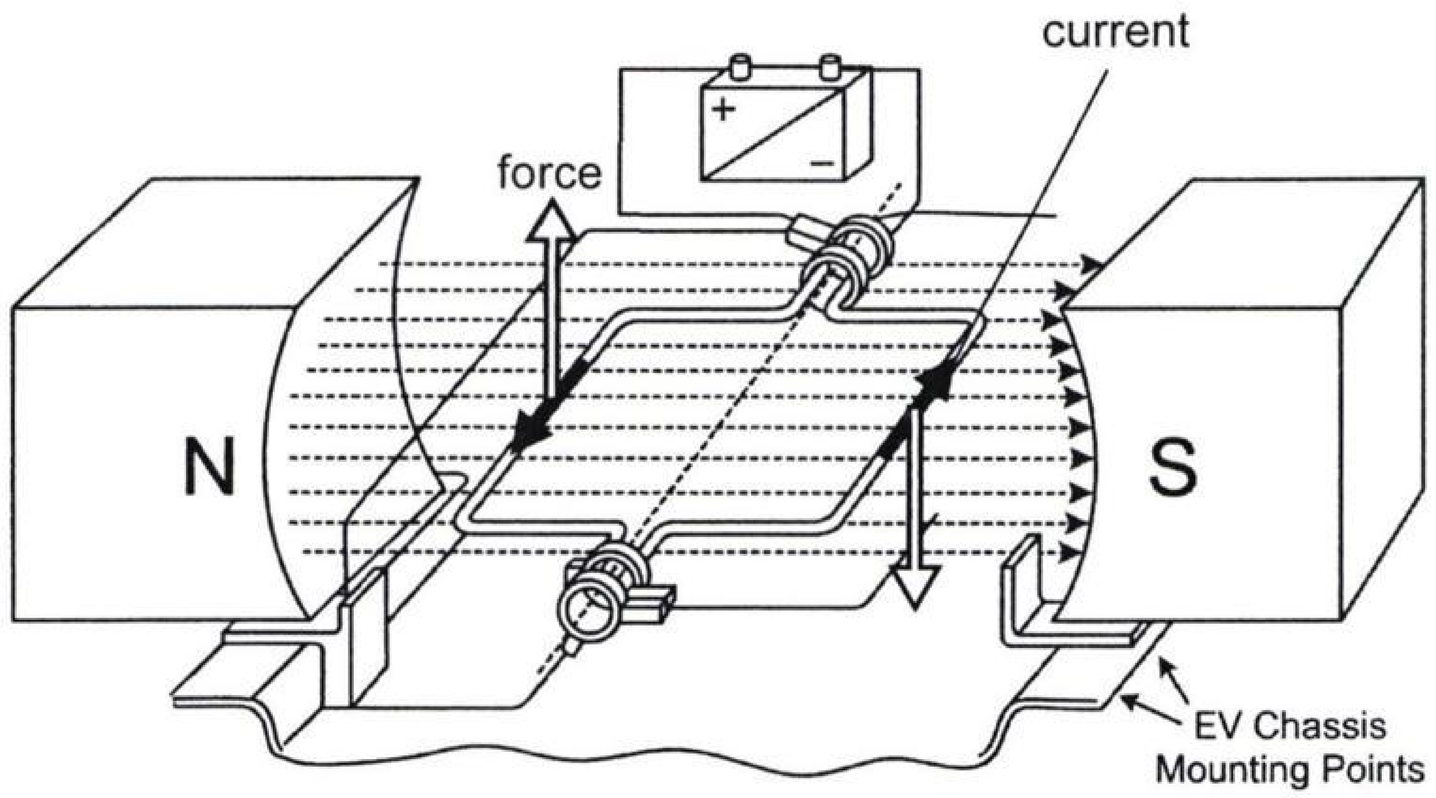


Fig. 8.1

Efficiency in an EV is further enhanced by regenerative braking. When the driver brakes, the motor's role is reversed and it acts as a generator, converting the vehicle's kinetic energy into electrical energy. As chemical batteries cannot absorb these rapid current surges without overheating, ultracapacitors are used as buffers.

Ultracapacitors are energy storage devices with very high capacitance and very low internal resistance. Since an individual ultracapacitor cell has a low voltage rating of 2.8 V, multiple cells are connected in series and parallel to form a high voltage ultracapacitor module needed by an EV.

Engineers tested a prototype ultracapacitor module with internal resistance, by charging it from a completely uncharged state with a constant current of 120 A from a power supply. Fig. 8.2 shows a simple circuit of the connection between the constant current power supply to the prototype ultracapacitor module.

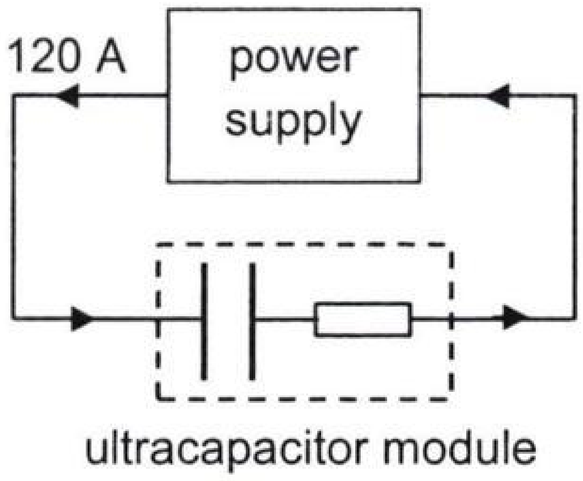


Fig. 8.2

The total voltage V_T across the ultracapacitor module was recorded over time t as shown in Fig. 8.3.

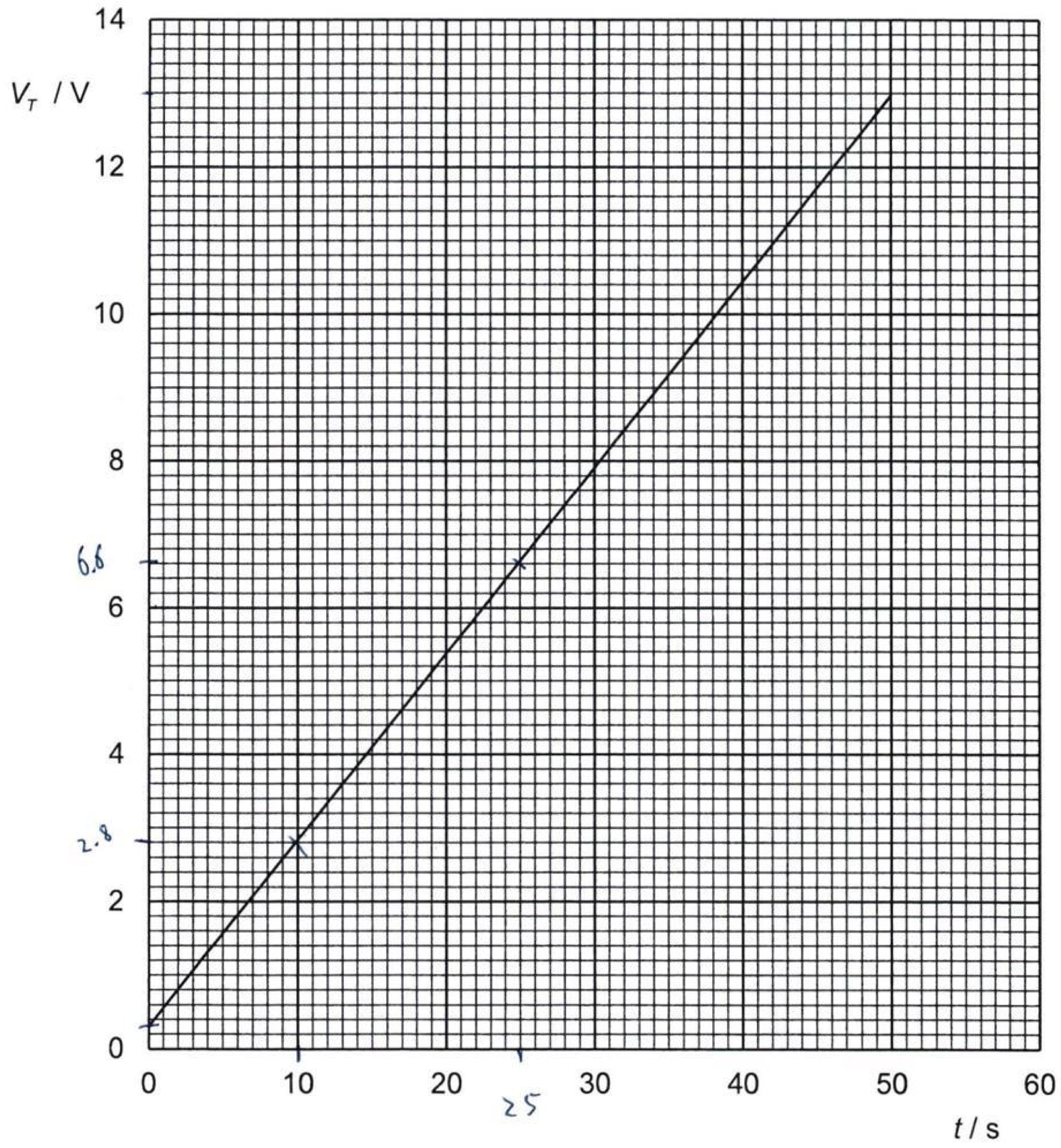


Fig. 8.3

(a) (i) For a petrol vehicle, calculate the average power required if the vehicle travels 100 km in one hour at constant speed.

1 / 2

work done = (7.0) (34.2 x 10^6)

time = 1 x 60 x 60 = 3600

power = $\frac{\Delta W}{\Delta t}$

= $\frac{(7.0)(34.2 \times 10^6)}{3600}$

= 665000 W

= 670 kW (2sf.)

power = ... 670 ..... kW [2]

(ii) The electricity for an EV is generated by a power plant operating at 50 % efficiency.

By comparing the energy per kilometer required by an EV and a petrol vehicle, explain why the EV is still considered 'greener'.

Support your explanation with appropriate calculations.

1 / 2

at 18 kWh x 2 = 36 kWh per 100km

while petrol vehicle is 670 kWh per 100km

eck

EV uses significantly less energy than petrol vehicle for same distance and same speed, so it is more energy efficient and does not waste as much as petrol vehicles.

..... [2]

(b) As the EV accelerates, the motor spins faster.

(i) Explain why the current in the motor decreases as the motor spins faster.

0 / 2

As the speed of the coil rotating in the magnetic field increases, the rate of change of magnetic flux in the coil increases as well, which induces a back e.m.f by Faraday's law of electromagnetic induction.

..... [2]

(ii) At a certain constant speed, the back e.m.f. is 300 V.

If the supply voltage is 360 V and the total coil resistance is 0.60 Ω , calculate the mechanical power developed by the motor.

0 / 2

voltage actually going through = 360 − 300 = 60V

$P = IV$

$= \frac{V^2}{R}$ ✗

$= \frac{(60)^2}{0.60}$

$= 600 = 0.60\text{kW}$

power = 0.60 ✗ kW [2]

(c) Using Fig. 8.3, determine for the ultracapacitor module,

(i) its internal resistance *r*,

1 / 1

~~at $t = 10$, $V = 2.8$~~ supply is $I = 120$

end charge is 13V at $t = 0$, $V = 0.3$

$V_0 = 13$

$2.8 = 13\left[1 - e^{-\frac{10}{I}}\right]$

$r = \frac{V}{I}$

$= \frac{0.3}{120}$ ✓

$= 0.0025$

r = 0.0025 ✓ Ω [1]

$\frac{2.8}{13} = e^{-\frac{10}{I}}$

$-10 = \ln\left(1 - \frac{2.8}{13}\right) = I$

(ii) the ideal voltage V_c across it at $t = 40$ s,

2 / 2

? Original voltage @ $t = 40$ is 10.4V.

in ideal cap, $r = 0$.

without resistance, P.D. drawn is 0.3V less

$V_c = 10.4 - 0.3$ ✓

$= 10.1$V

V_c = 10.1 ✓ V [2]

(iii) its capacitance *C*.

0 / 2

$V_0 = 13$

$2.8 = 13\left[1 - e^{-\frac{10}{I}}\right]$

$1 - \frac{2.8}{13} = e^{-\frac{10}{I}}$

$\frac{-10}{\ln\left(1 - \frac{2.8}{13}\right)} = I = R($

$\left(= \frac{-10}{(0.0025)\ln\left(1 - \frac{2.8}{13}\right)}\right.$

$= 1640.7$ ✗

$= 16000$ F (2 s.f.)

C = 16000 ✗ F [2]

(d) Each individual ultracapacitor cell has a capacitance of 3600 F and a voltage rating of 2.8 V.

(i) Determine the number of cells N that must be connected in series to support a voltage of 42 V across this series string.

1 / 1

voltage is split

$\frac{42}{2.8} = 15$

N = ... 15 ..... [1]

(ii) Calculate the effective capacitance of N cells in the series string.

1 / 1

$\left(\frac{1}{3600} \times 15\right)^{-1} = 240$

capacitance = ... 240 ..... F [1]

(iii) An ultracapacitor module consists of an arrangement of series strings each of N cells.

Compare your answers in **(c)(iii)** and in **(d)(ii)**,

State, with a reason, what can be deduced about the arrangement of the series strings of cells in the module.

1 / 1

It is a group of parallel series strings. Capacitor adds up in parallel, and putting 240F series strings in parallel adds to larger capacitance. [1]

(e) To test an emergency roadside charging system, engineers connected an uncharged ultracapacitor module across a constant 24 V DC power supply in series with the vehicle's climate control system, which acts as a constant resistive load of $0.50\ \Omega$.

Together, the ultracapacitor module and the climate control system form a simple RC charging circuit. The internal resistance of the ultracapacitor module is neglected during this test.

The variation of the voltage V across the ultracapacitor module with time t during this charging process is given by

$$V = V_0\left(1 - e^{-t/\tau}\right)$$

where is τ is the time constant of the RC circuit.

Use your value of C from **(c)(iii)** for the ultracapacitor module in the following calculations.

(i) Calculate the value of τ for the emergency charging circuit.

0 / 1

τ = ... s [1] 0

(ii) On the axes in Fig. 8.4, sketch a graph showing how the voltage V across the ultracapacitor module varies with time t from $t = 0$ to $t = 5\tau$.

Label the numerical value of V at $t = \tau$ and any other key numerical values on the vertical axis.

0 / 2

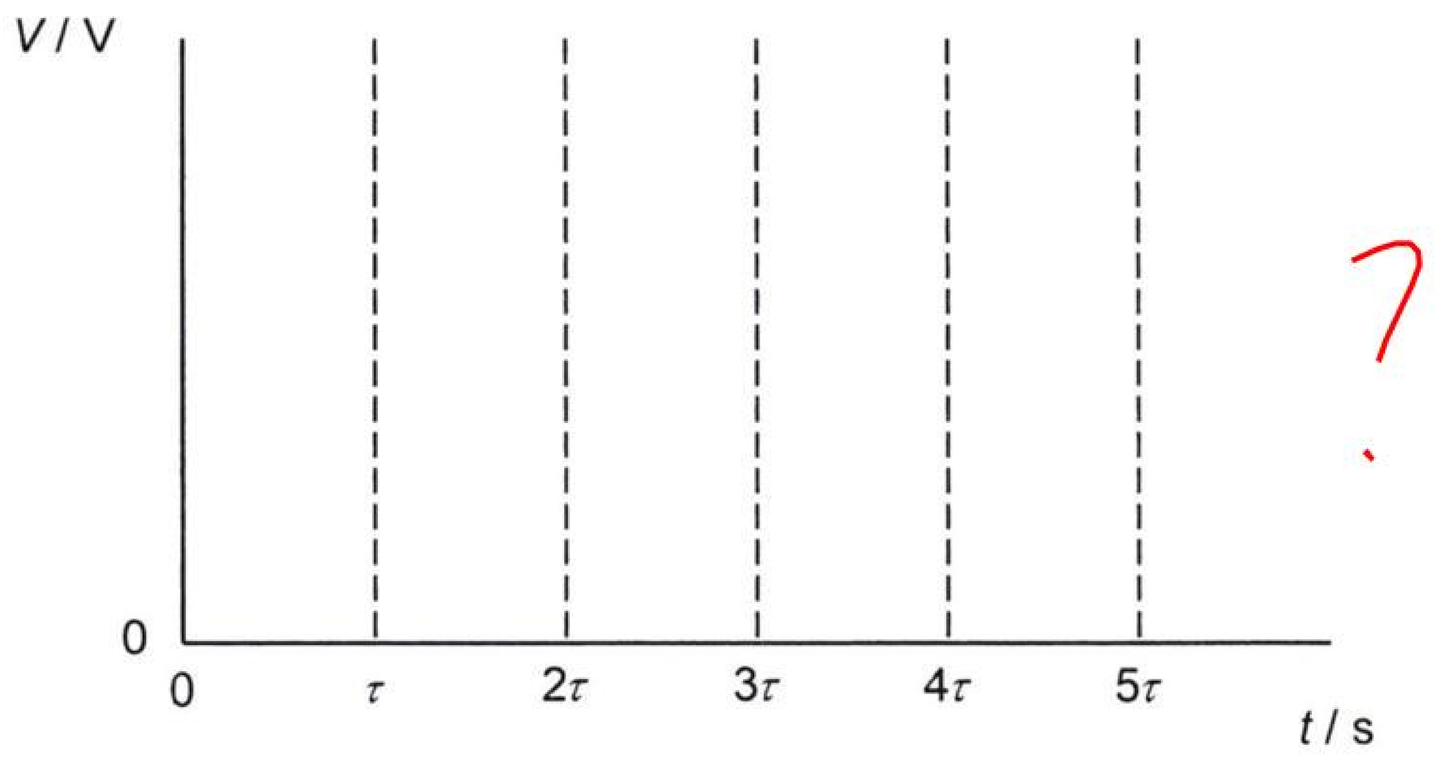


Fig. 8.4

[2] 0

(iii) The ultracapacitor module is allowed to charge for a long time until it is fully charged to 24 V.

1. Calculate the total electrical energy stored in the fully charged module.

0 / 1

energy = ... J [1] 0

2. Calculate the total energy delivered by the 24 V DC supply.

0 / 1

energy = ... J [1] 0

3. State a reason for the difference in the energy values calculated in **(e)(iii)1.** and **(e)(iii)2.**

0 / 1

..... [1] 0

[Total: 22]

how many times has this been tested?

5 A capacitor of capacitance C is connected in parallel with a resistor of resistance 200 Ω across an alternating supply as shown in Fig. 5.1.

$C = \frac{Q}{V}$

$U = \frac{1}{2}QV$

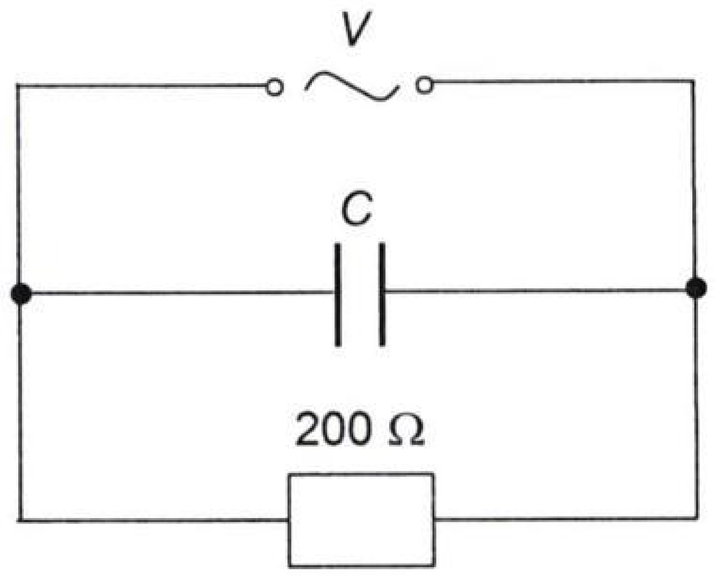


Fig. 5.1

The variation with time t of the potential difference V of the alternating supply is given by

$$V = 120\sin(314t),$$

where V is measured in volts and t in seconds.

(a) (i) Write an equation, in terms of C, that relates how charge Q on the capacitor plate varies with t.

1 / 1

$C = \frac{Q}{V}$

$Q = CV$

$= (120\sin(314t))$ ✓

[1]

(ii) Hence, show the current I_C through the capacitor is $37700C\cos(314t)$.

1 / 1

$I_C = \frac{dQ}{dt}$

$= (120)(314)(\cos(314t))$

$= 37680(\cos(314t))$ ✓

$= 37700(\cos(314t))$ (3 s.f.)

[1]

(b) (i) Determine the r.m.s. current $I_{R,rms}$ through the resistor.

1 / 1

$I_{krms} = \frac{V_{rms}}{R} = \frac{\frac{\sqrt{2}}{2}(120)}{200}$

$= 0.42426$

$= 0.424$ A (3 s.f.)

$I_{R,rms}$ = 0.424 ✓ A [1]

(ii) Calculate the mean power P dissipated in the resistor.

0 / 2

$P = I^2R$

~~IAV~~

$\langle P \rangle = I_{rms}^2 R$

$= (0.42426^2)(200)$

$= 84.852$

$= 84.9\ W$ (3 s.f.)

$P =$ 84.9 W [2]

(c) The frequency of V is increased.

State and explain the effect, if any, on

(i) the r.m.s. current through the resistor,

1 / 1

No effect. r.m.s voltage is dependent on amplitude and shape of alternating current, not frequency, so current which is $\frac{V_{rms}}{R}$, has no change. [1]

(ii) the r.m.s. current through the capacitor.

0 / 1

[1]

[Total: 7]

6 Fig. 6.1 shows the current-voltage (I-V) characteristics of a resistor and a diode.

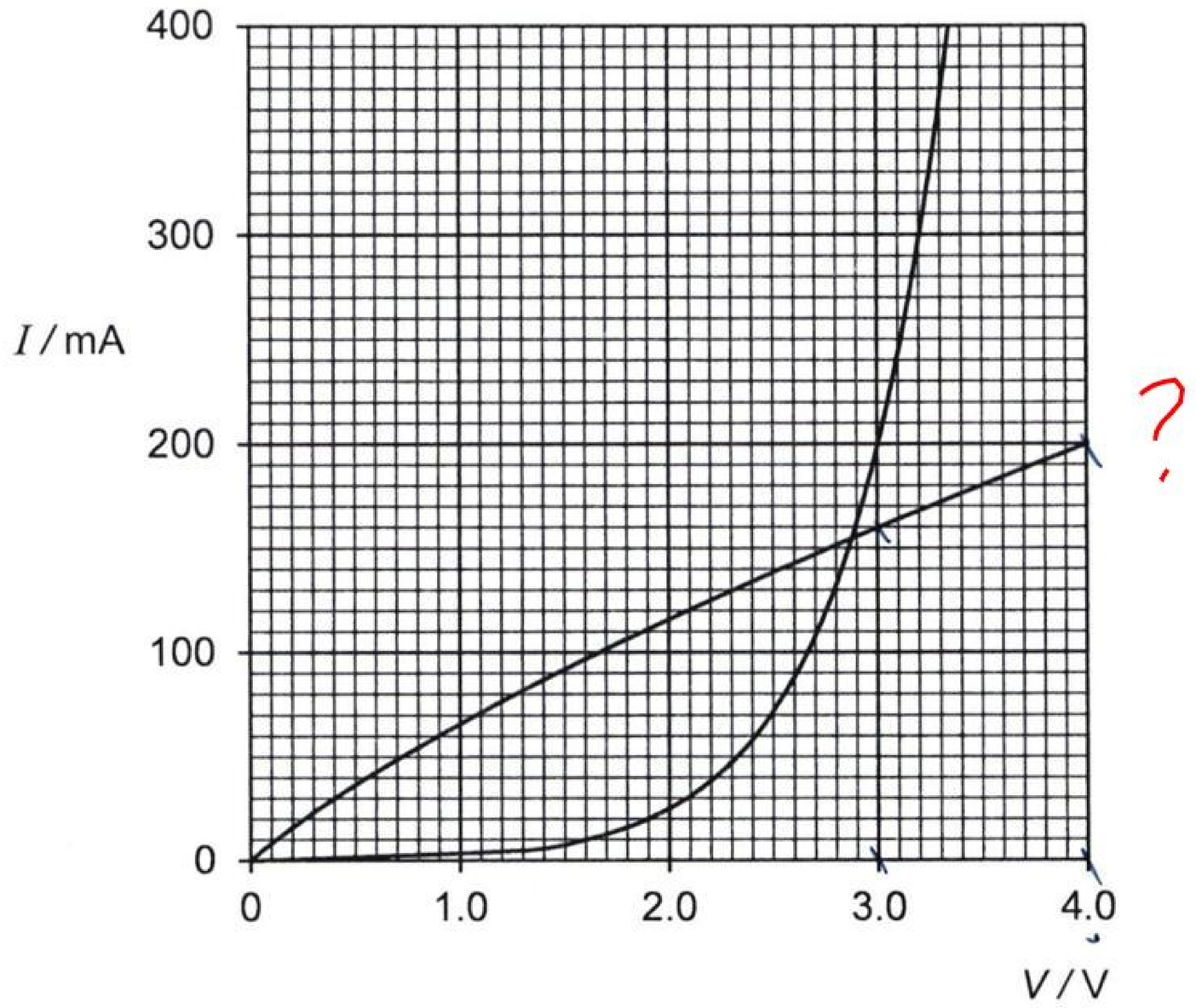


Fig. 6.1

When the resistor and diode are connected in series with a battery of negligible internal resistance, the current in the circuit is 200 mA.

(a) Determine

(i) the e.m.f. of the battery,

@ 200 mA

2 / 2

P.D. across both components = e.m.f. = 3.0 + 4.0
= 7.0 V
– 2 dp.

e.m.f. = 7.0 V [2]

sf/dp

0 / 1

(ii) the resistance of the diode.

$V = IR$

$R = \frac{V}{I}$

$= \frac{4.0}{40200 \times 10^{-3}}$ ✗

$= 20\ \Omega$

resistance = ... 20 ✗ ..... Ω [1]

(b) (i) On Fig. 6.1, sketch the *I-V* characteristic of a negative temperature coefficient thermistor. Exact values are not required. [1]

0 / 1

Not drawn.

(ii) Explain, in microscopic terms, the difference in the *I-V* characteristics between the thermistor and the resistor.

0 / 1

.....

.....

..... [1]

[Total: 5]

7 A proton travelling in a vacuum at a speed of 2.0×10^5 m s^{-1} enters a region of uniform magnetic field of magnetic flux density 3.5 mT . The magnetic field is directed into the plane of the paper.

Initially, the proton is travelling in a direction normal to the magnetic field. The path of the proton while in the magnetic field is the arc of a quarter circle as shown in Fig. 7.1.

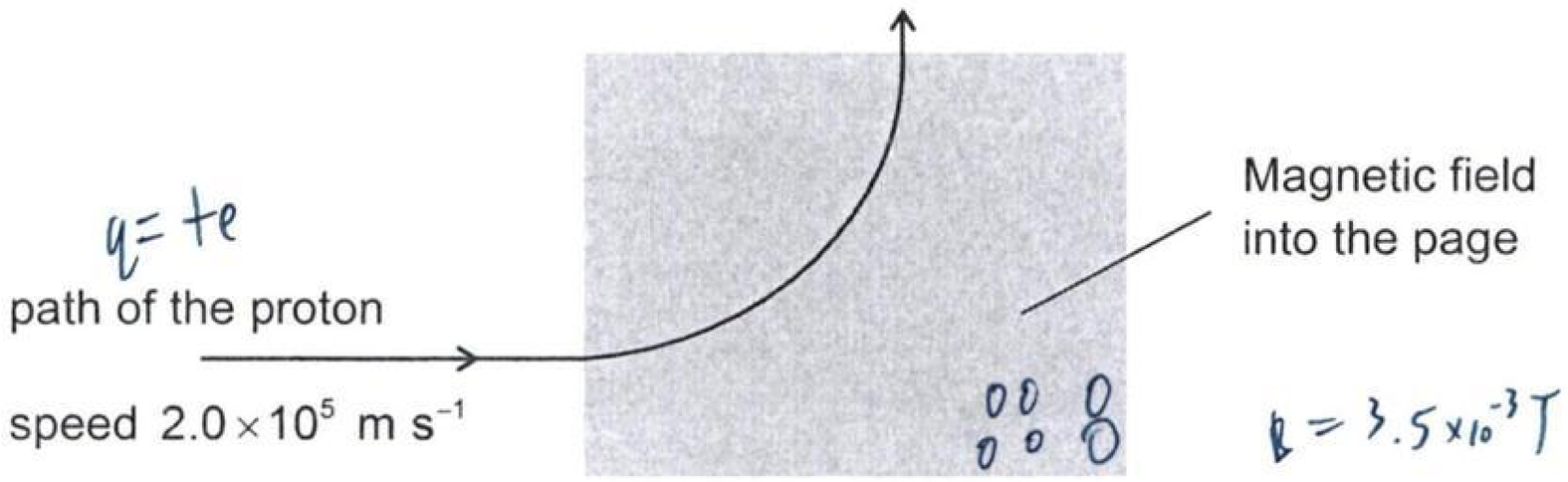


Fig. 7.1

(a) Explain why the path in the magnetic field is an arc of a circle.

0 / 2

Using fleming's left hand rule, we can deduce that the path of the proton that is normal to the magnetic field produces a magnetic force acting on it perpendicular to both the magnetic field and path of proton. The force that is perpendicular to the motion of the proton results in the circular motion of the proton, causing it to arc in a quarter circle until exit fon. [2]

missing key points (-2)

(b) (i) Show that the radius of the circular arc is 0.60 m .

1 / 1

$F = Bqv$ provides centripetal force $\frac{mv^2}{r}$

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq}$$

$$= \frac{(1.67\times10^{-27})(2.0\times10^5)}{(3.5\times10^{-3})(1.60\times10^{-19})}$$

$$= 0.5964$$

$$= 0.60\,\text{m (2 s.f.) (shown)}$$

[1]

(ii) By considering the change in momentum of the proton, calculate the magnitude of the average force exerted on the proton while it was in the uniform magnetic field.

4 / 4

by newton's second law, $F = \frac{\Delta p}{\Delta t}$

mass is the same, so $\Delta p = m \Delta V$

in circular motion,



$\Delta V = 2V \sin\frac{\Delta\theta}{2}$

$\Delta p = 2mV \sin\frac{\Delta\theta}{2}$

rate of change in angle $\frac{\Delta\theta}{\Delta t} = \omega = \frac{V}{r}$

in quarter circle, $\Delta\theta = 90^0$

$\Delta p = 2mV \sin\frac{90^0}{2}$

$= \sqrt{2}\, mV$

$\Delta t = \frac{\text{distance}}{V} = \frac{\frac{1}{2}\pi r}{V} = \frac{\frac{1}{2}(\pi)(0.[illegible]5464)}{2.0 \times 10^5}$

$= 4.684 \times 10^{-6}$ s

$F = \frac{\sqrt{2}\, mV}{4.684 \times 10^{-6}} = \frac{\sqrt{2}(1.67 \times 10^{-27})(2.0 \times 10^5)}{4.684 \times 10^{-6}}$

$= 1.008 \times 10^{-16}$

$= 1.0 \times 10^{-16}$ N

average force = 1.0 x 10^-16 N [4]

[Total: 7]

End of Section A

Section B

Answer **one** question in this section in the spaces provided.

8 **(a)** Define *magnetic flux density*.

0 / 1

.....

.....

..... [1]

(b) **(i)** Fig. 8.1 shows a flat circular coil carrying a current in the direction shown.

1 / 3

Sketch the magnetic field pattern due to the current in the coil within the dashed region.

more than 2 parallel lines in the middle

lines are not equally spaced

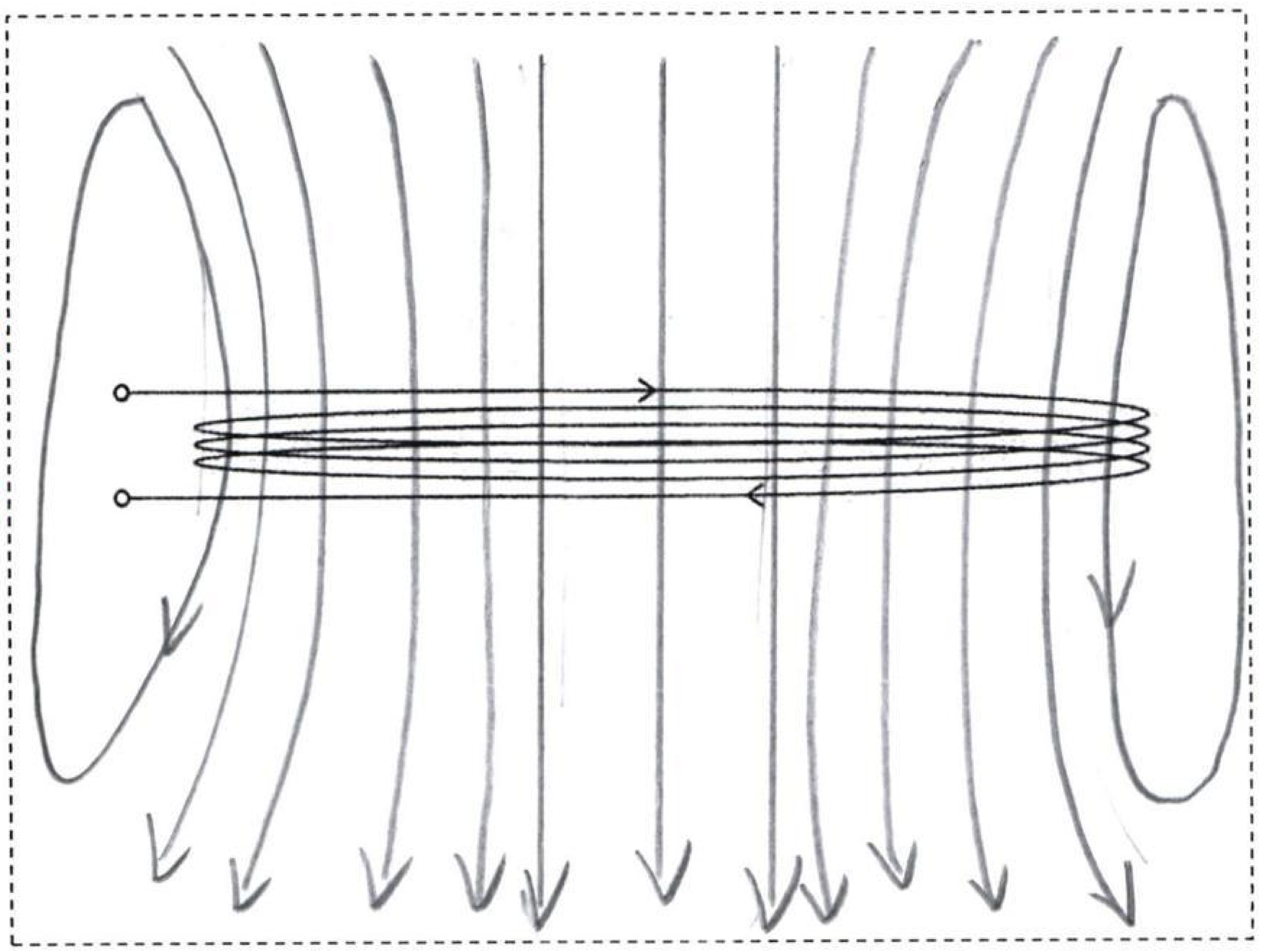


Fig. 8.1 (not to scale)

[3]

(ii) The flat circular coil in **(b)(i)** is fixed in position. A straight wire carrying current is then placed near the flat circular coil.

Explain the effect on the straight wire when it is placed

1. above the coil as shown in Fig. 8.2,

2 / 2

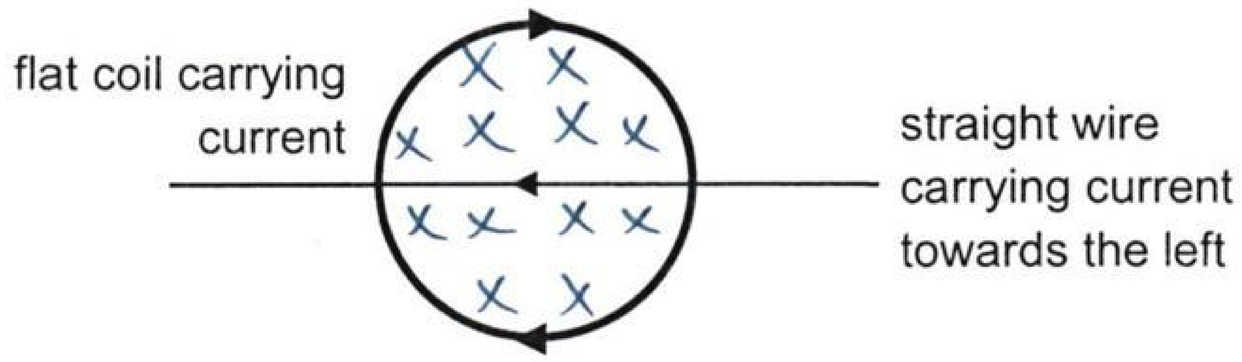


Fig. 8.2 (top view)

Wire will move downwards. Using flaming's left hand rule, force on wire points downwards from the magnetic field pointing into the page derived with right hand grip rule, and the current of wire flowing towards the left.

2. along the axis of the coil, as shown in Fig 8.3.

1 / 1

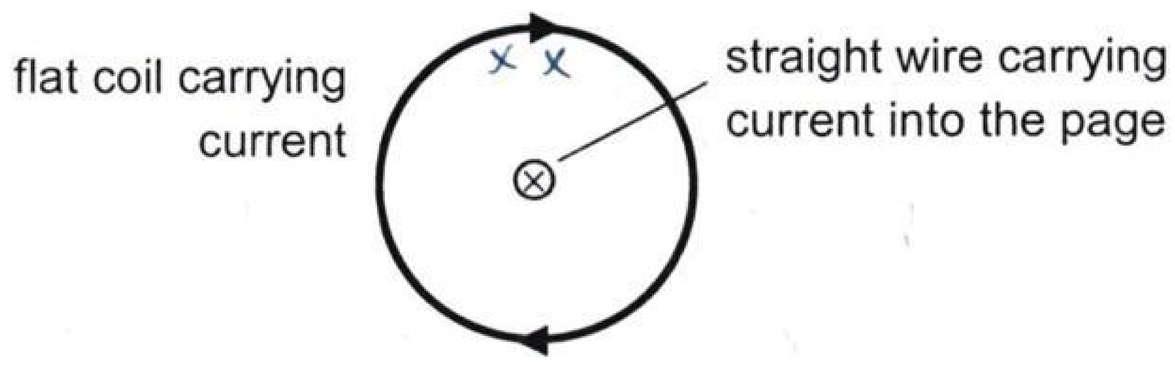


Fig. 8.3 (top view)

No effect. Current is parallel to magnetic field, which results in no component of both being perpendicular to each other and there resulting in 0 force on the wire.

[3]

(c) Fig. 8.4 shows a metal saucepan placed on an electromagnetic induction cooking hob.

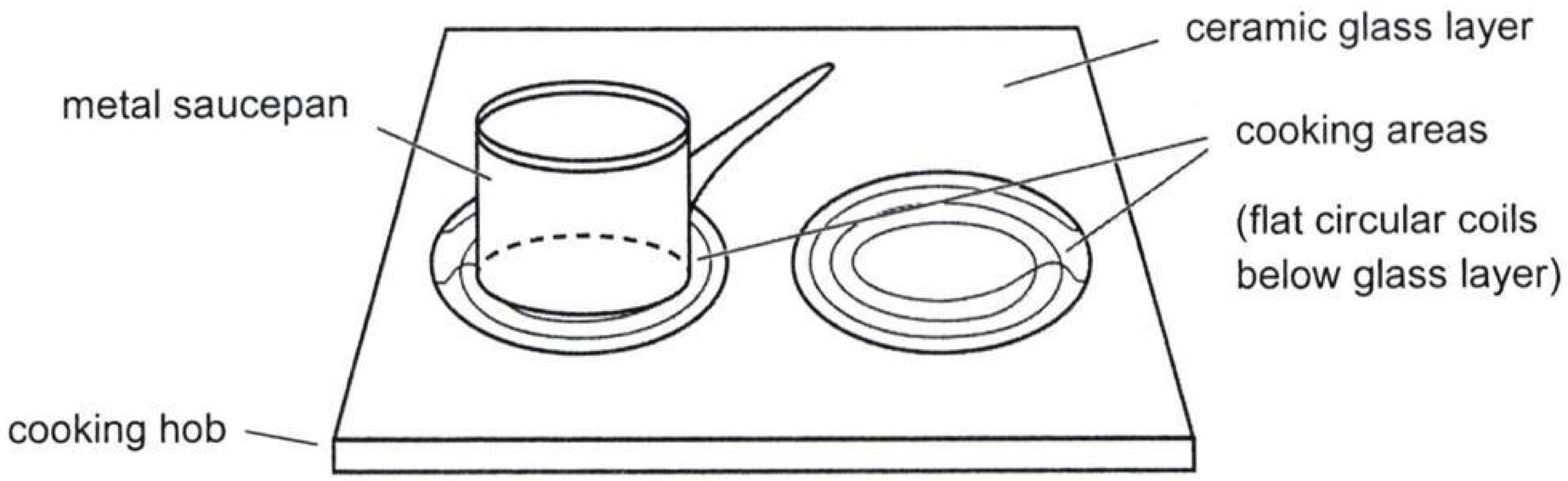


Fig. 8.4

Each cooking area of the induction cooking hob consists of a flat circular coil with 20 turns below a layer of ceramic glass. When switched on, the coils carry a high frequency alternating current.

(i) Explain how a high frequency alternating current in the coils results in a significant transfer of energy to the contents in the saucepan.

3 / 3

By Faraday's law of electromagnetic induction, the induced e.m.f. is ~~the~~ proportionate to rate of change in magnetic flux linkage. When the coils alternate in current at a high frequency, the magnetic ~~flux linkage of the~~ ~~coils~~ fields of the coils constantly alternates at the same frequency. This produces a rapidly changing magnetic field, ~~it~~ which in turn rapidly changes the magnetic flux linkage of the saucepan. The ~~w~~high rate of change in magnetic flux linkage induces a large, significant amount of eddy currents in the saucepan, which ends up transferring the heat ~~lost in the~~ from the eddy currents to the contents in the saucepan. [3]

emf?

(ii) Fig. 8.5 shows the variation with time t of the alternating current I_{coil} flowing through the coil.

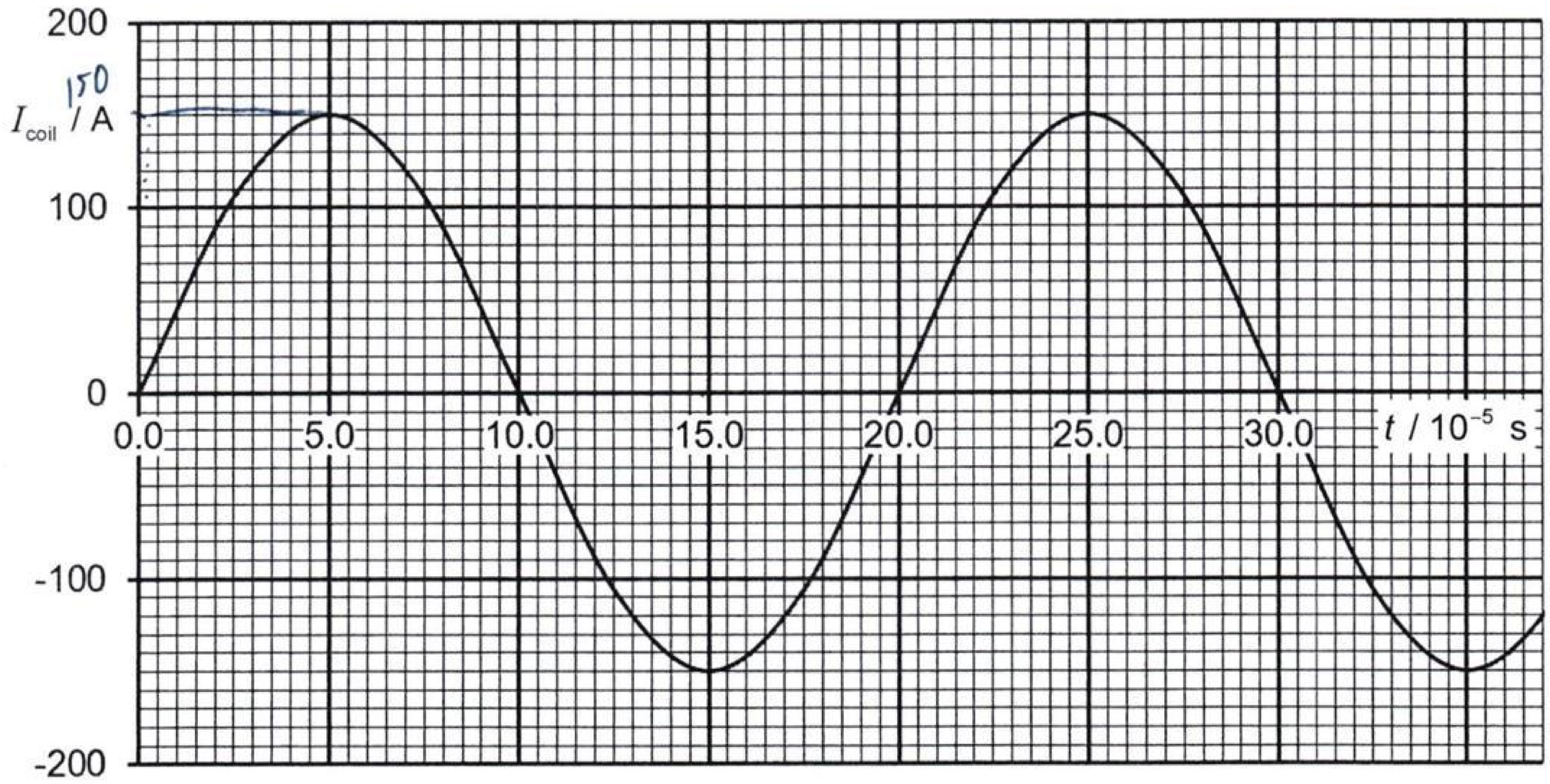


Fig. 8.5

The metal saucepan and its contents have a total mass of 900 g and an effective specific heat capacity of 2000 J kg^{-1} K^{-1}. The circular base of the saucepan has an effective area and resistance of 0.020 m^2 and 0.20 Ω respectively.

The magnetic flux density throughout the base of the saucepan can be taken to be $5\mu_o NI_{coil}$ where μ_o is the permeability of free space and N is the number of turns in the flat circular coil.

Assume there is no loss of energy between the coil and the saucepan.

1. Use Fig 8.5 to determine the maximum value of the induced current in the base of the saucepan.

1 / 3

amplitude = 150
period = 20 s

$I_c = 150 \sin\left(\frac{2\pi}{20}t\right)$

$\frac{dI_c}{dt} = 3000\pi \cos\left(\frac{2\pi}{20}t\right)$

$\left.\frac{dI_c}{dt}\right|_{t=10} = 3000 \cos\left(\frac{2\pi}{20} \times 10\right)$

$= 2995$

$= 3000$

induced emf$_{max} = \frac{\Delta\Phi}{\Delta t} = \frac{dB}{dt} A$

$= \frac{dI}{dt} (5\mu_0 N A)$

$= (3000)(4\pi \times 10^{-7})(5)(4\pi \times 10^{-7})(0.020)(20)$

$= 7.5340 \times 10^{-3}$

max induced current $= \frac{V}{R} = \frac{7.540 \times 10^{-3}}{0.2}$

$= 0.03770$

$= 0.038$

maximum value of induced current = ... 0.038 ..... A [3]

0 / 3

2. Sketch, on Fig. 8.6, a graph to show the variation with t of the ~~rate of transfer of energy from the coil to~~ the saucepan for the first 30×10^{-5} s. Label all relevant values.

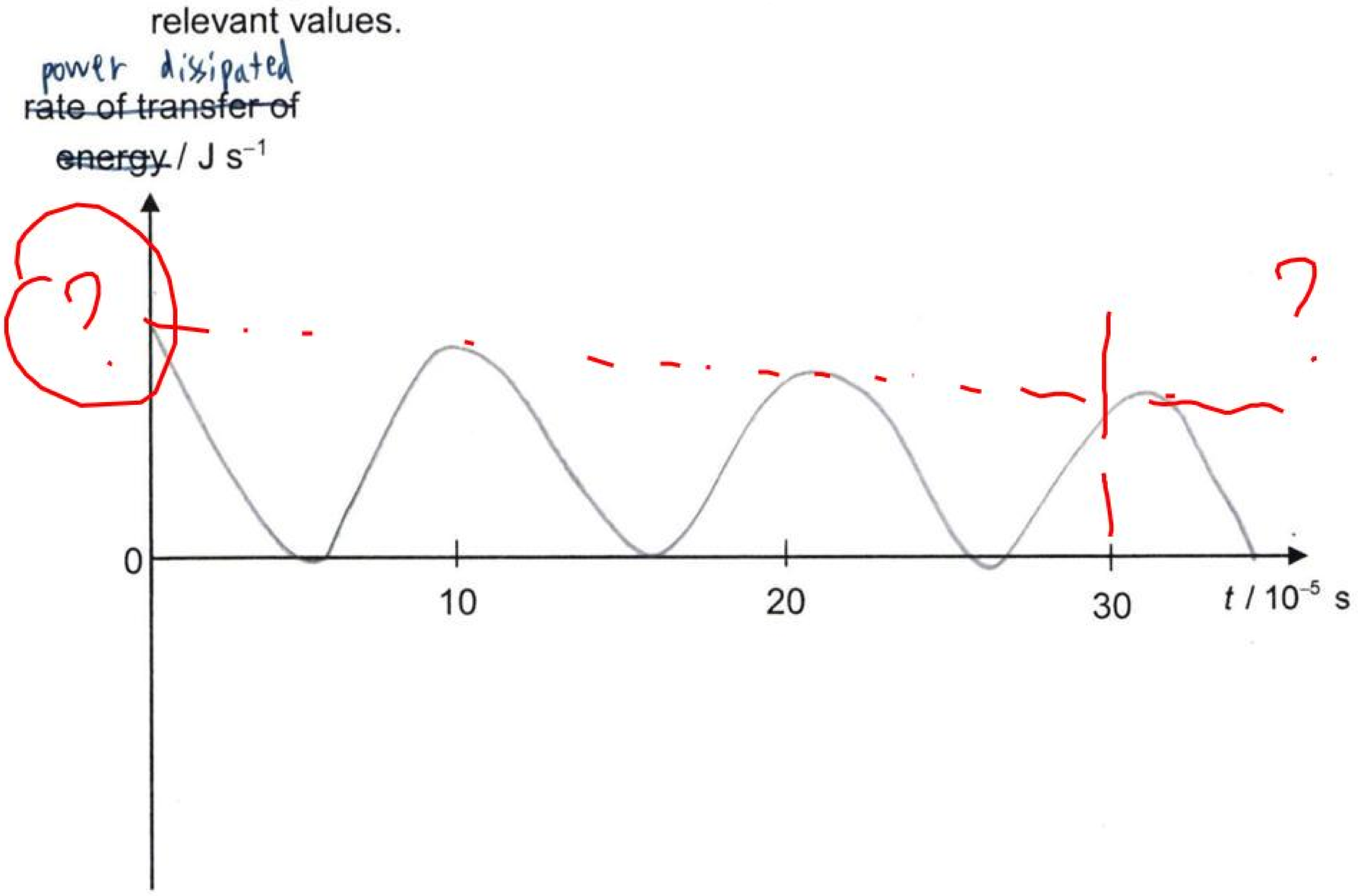


Fig. 8.6

[3]

3. Calculate the temperature rise of the saucepan and contents after 4.0 minutes of use.

0 / 3

temperature rise = ... ° C [3]

(iii) Stainless steel has a high magnetic permeability. Explain why stainless steel cookware are preferred in induction cooking.

0 / 1

This allows for more efficient and faster heat transfer. Why?

.....

..... [1]

[Total: 20]

9 **(a)** State how a stationary wave is formed.

.....

.....

.....

..... [2]

(b) A stationary wave is set up in a stretched string. The variation of the maximum vertical displacement *y* of the string with distance *x* along the stationary wave is shown in Fig. 9.1.

P, Q and R are points on the string. The time taken by point P to travel from its maximum upward displacement to its maximum downward displacement is 9.0×10^{-3} s.

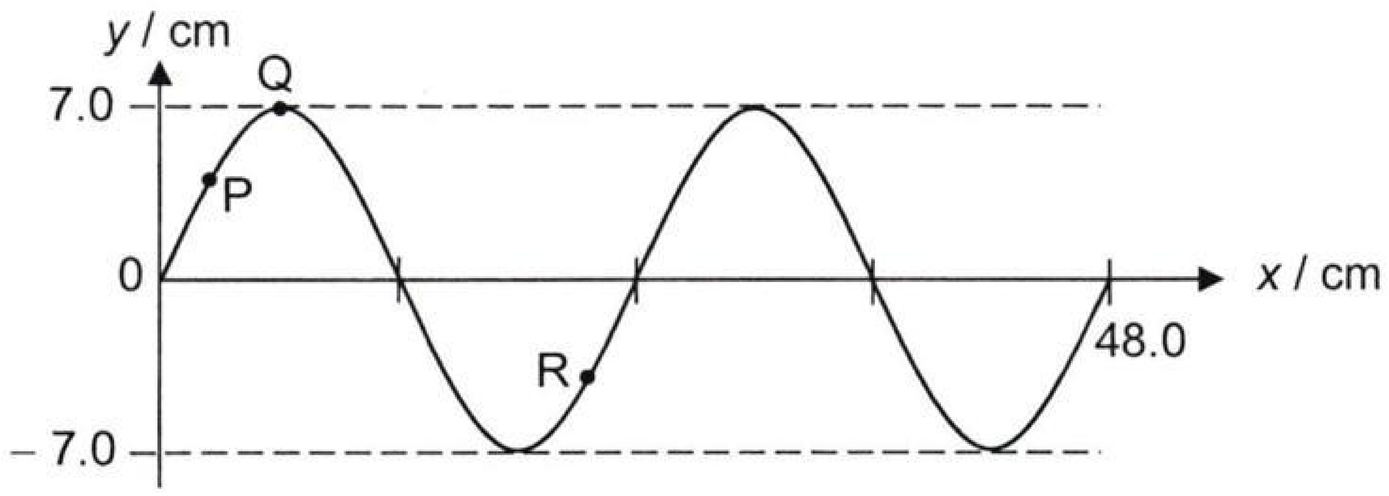


Fig. 9.1

(i) Determine, for each wave which formed the stationary wave, its

1. amplitude,

amplitude = ... cm [1]

2. frequency,

frequency = ... Hz [1]

3. speed.

speed = ... m s^{-1} [2]

(ii) State the phase difference between points

1. P and Q,

phase difference = ... rad [1]

2. Q and R.

phase difference = ... rad [1]

(c) A particle of mass *m* is trapped in a one-dimensional box of width *L* with perfectly reflecting walls. Its wavefunction is given by

$$\Psi_n(x) = \begin{cases} A\sin\left(\dfrac{2\pi x}{\lambda_n}\right) & 0 \le x \le L \\ 0 & \text{otherwise} \end{cases}$$

where λ_n is the de Broglie wavelength of the particle, *n* is the quantum number and *A* is the normalisation constant.

(i) Explain why the value of $\int_0^L |\Psi|^2 \, dx$ must be 1.

.....

.....

..... [1]

(ii) Write down an expression for λ_n in terms of L and n.

[1]

(iii) For $n = 3$, the wavefunction is $\Psi_3(x)$.

On Fig. 9.2, sketch the probability density function for the particle. Label the axes with appropriate values.

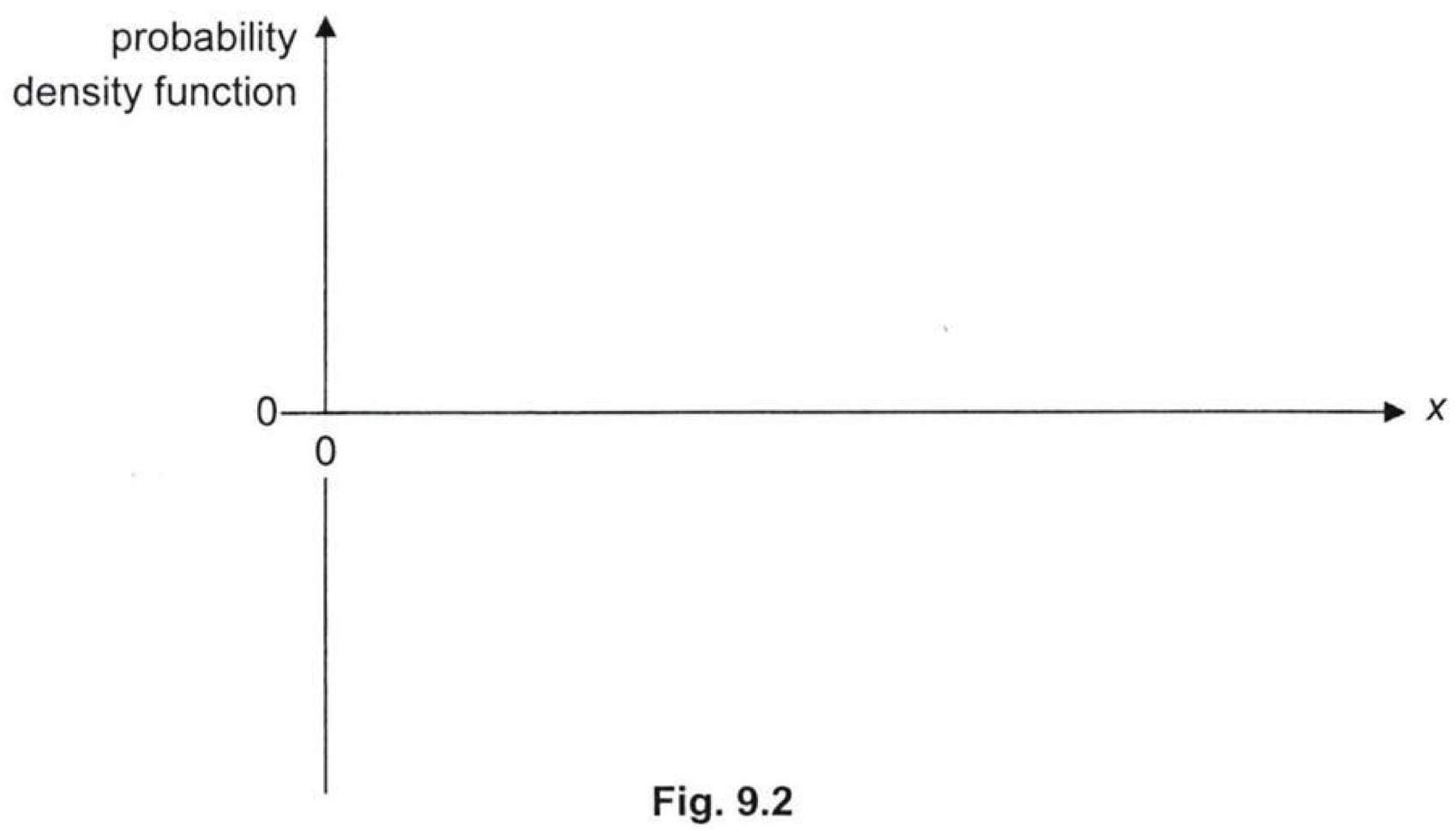


Fig. 9.2

[2]

(iv) Calculate the value of A. Show your working clearly.

[2]

(v) Suggest what might happen to the particle if the walls of the box were not perfectly reflecting.

.....

.....

..... [1]

(d) In 1980, Japanese Physicist Akira Tonomura carried out a double-slit experiment using electrons.

Fig. 9.3 shows the number of electrons that have arrived at the screen for various elapsed times *t*.

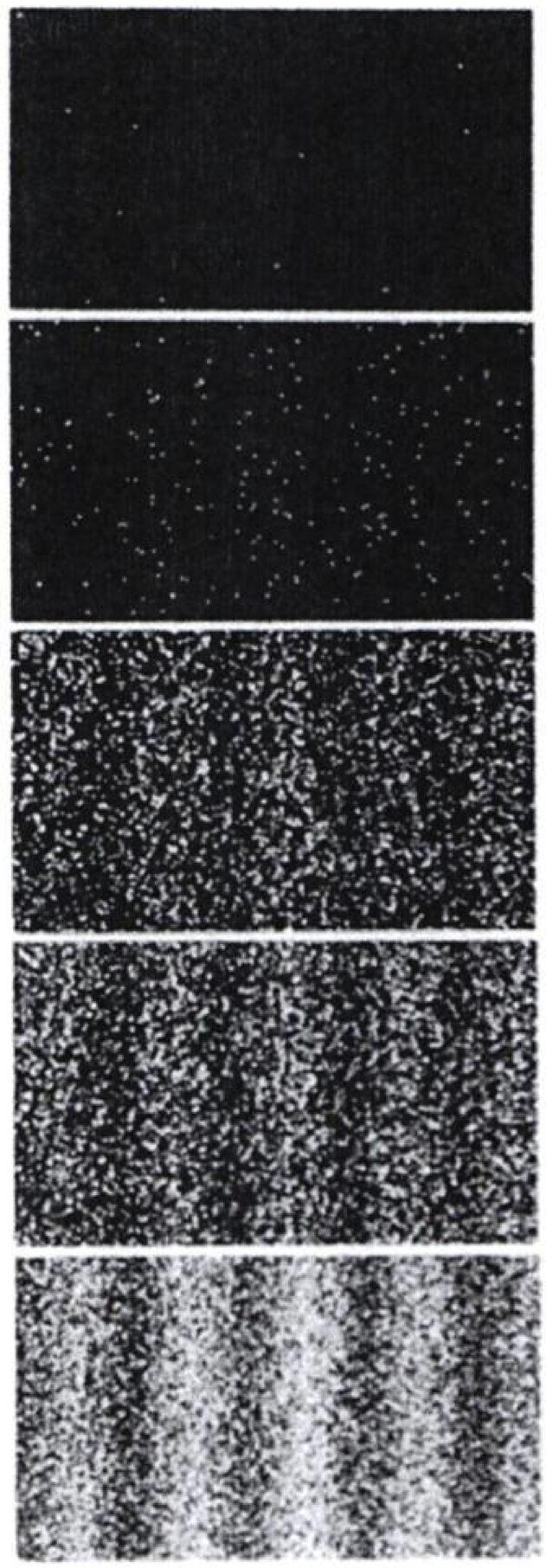


Fig. 9.3

4 A technician is designing a charging circuit for a compact camera flash unit as shown in Fig. 4.1. The circuit consists of a DC battery V of 12 V, a resistor R = 15 kΩ and an uncharged capacitor C of 220 μF, connected in series with a switch.

At time t = 0, the switch is closed.

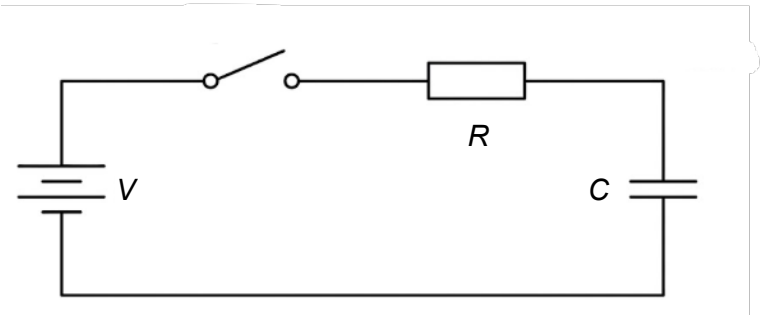


Fig. 4.1

(a) Calculate the time constant for the circuit.

time constant = ... s [1]

(b) Calculate the percentage of maximum charge in the capacitor 6.6 s after the switch is closed.

percentage = ... % [2]

(c) Hence or otherwise, determine the current flowing through the circuit exactly 6.6 s after the switch is closed.

current = ... mA [2]

[Total: 5]

5 A circuit is set up as shown in Fig. 5.1. The circuit contains a battery of e.m.f. 12 V, and a fixed resistor r of resistance 6 Ω and length L_r = 0.50 m. A movable metal rod M, of length L_M = 0.60 m and resistance 12 Ω, is placed on a conducting track at a distance d = 4.0 cm from r. The conducting track is assumed to be smooth and has zero resistance.

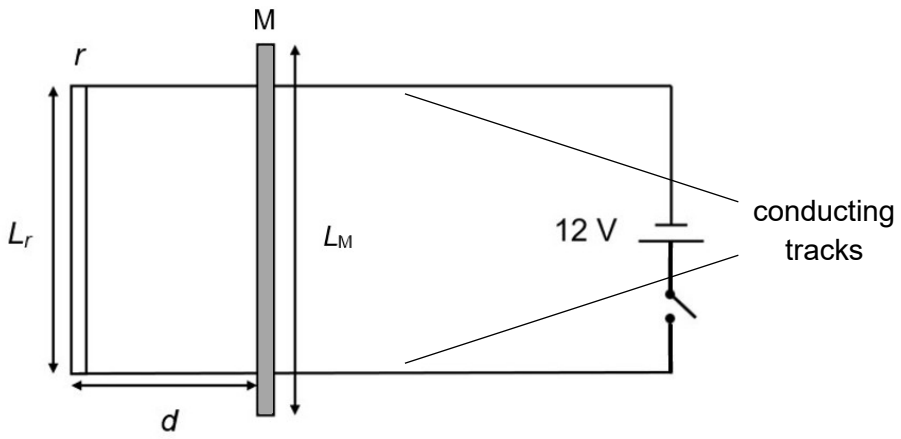


Fig. 5.1 (not to scale)

The switch is now closed.

(a) Show that the magnetic flux density B at the metal rod M due to the current in resistor r is 10 μT.

[1]

(b) Calculate the magnetic force experienced by M and state its direction.

magnetic force = ... N

direction: ..... [3]

(c) M in **(b)** is held stationary and a magnetic field of flux density 1.0 mT is now directed out of page throughout the circuit in Fig. 5.1. M is now released.

Describe and explain the initial motion of M.

……..

……..

……..

……..

…… [2]

(d) A proton enters the magnetic field *B* at an angle to the field of less than 90°.

The proton follows the path shown in Fig. 5.2.

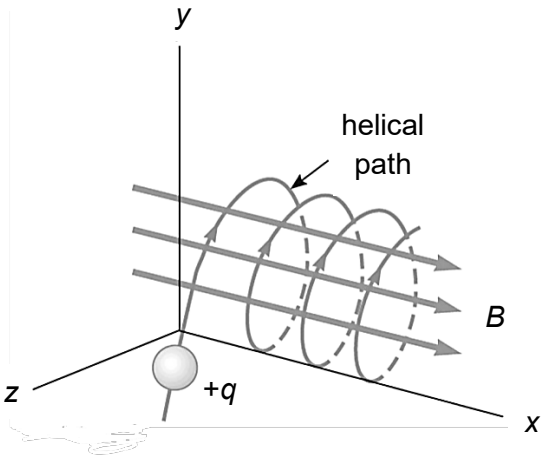


Fig. 5.2

(i) Explain why the proton follows this helical path.

……

……

……

……

……

……

…… [2]

(ii) Fig. 5.3 shows a magnetic bottle, where a non-uniform magnetic field is produced between two current carrying coils. A proton is initially directed perpendicularly into the page with speed v at the position shown.

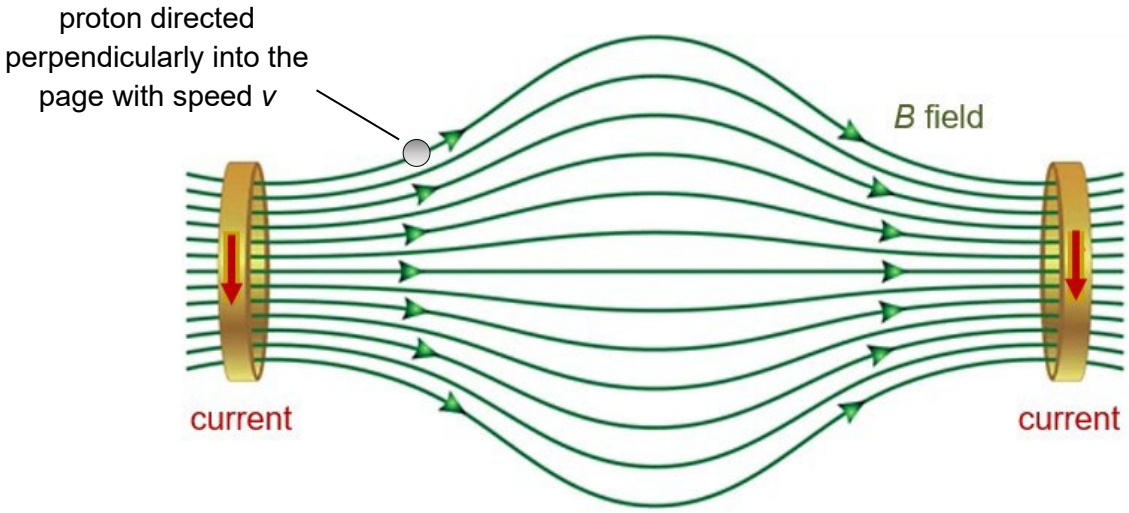


Fig. 5.3

Describe and explain the subsequent motion of the proton in the magnetic bottle, assuming the proton remains within the field region throughout.

……

……

……

……

……

……

……

……

……

……

…… [2]

[Total: 10]

7 Read the passage below and answer the questions that follow.

Traditional wind turbines are limited by their height; the friction of the Earth's surface slows wind speeds at low altitudes, and the massive concrete foundations required for towers over 150 metres tall are both expensive and environmentally invasive. To overcome these limitations, engineers have developed **Airborne Wind Energy Systems (AWES)**.

In early 2026, the S2000 Stratosphere Airborne Wind Energy System (SAWES), developed in China, became the world's first megawatt-class "flying power plant" to successfully feed electricity into a national grid. The system utilizes a helium-filled, ring-shaped aerostat (a "balloon") anchored to a ground station by a high-strength tether. A simplified schematic of the S2000 system is shown in Fig. 7.1. The aerostat remains aloft through upthrust, the upward force exerted by the surrounding air on the balloon when it is immersed in the air. This allows the turbines to access the "jet effect" of the stratosphere—a region where winds are faster, more stable, and carry significantly more kinetic energy than those at ground level.

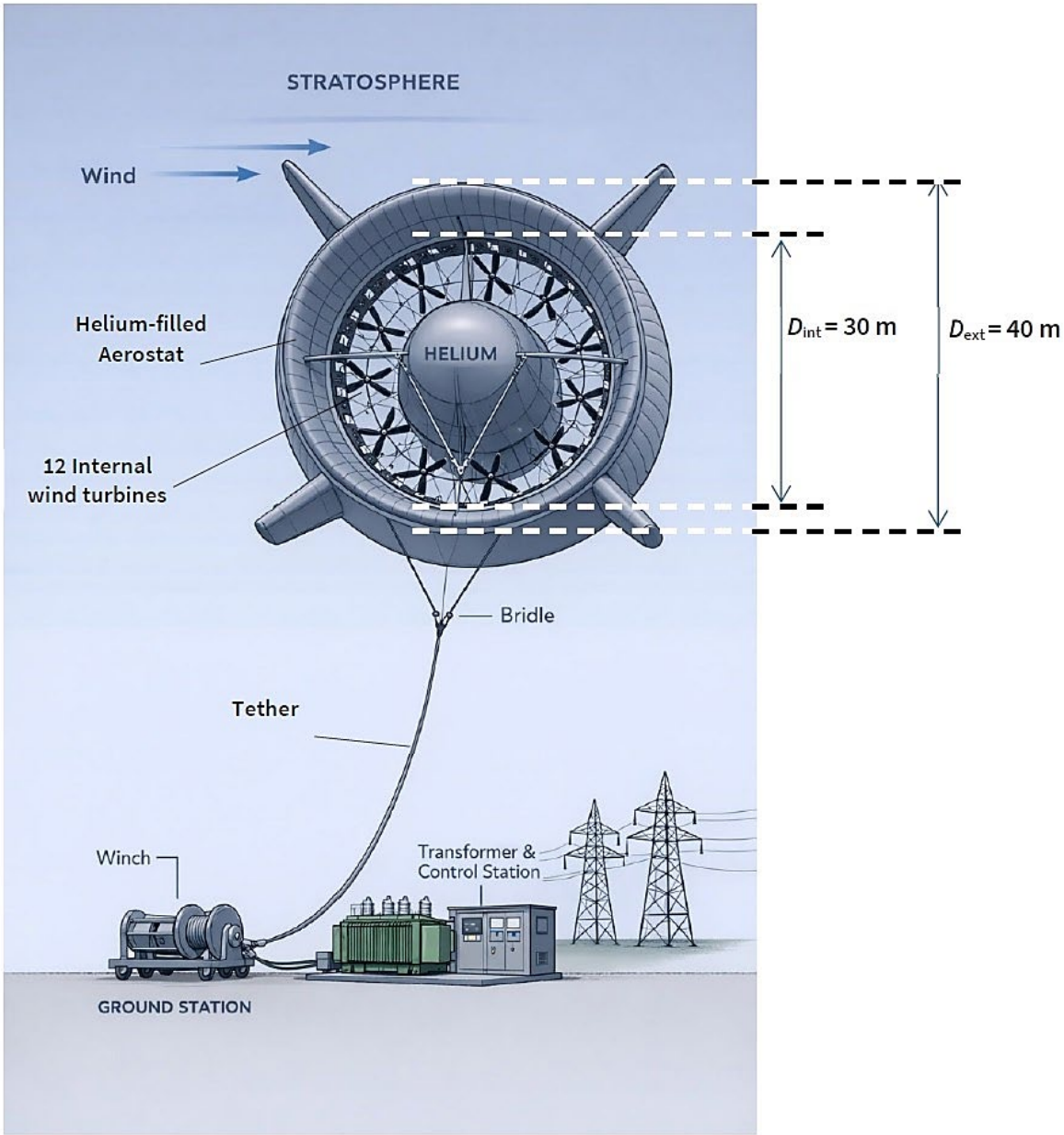


Fig. 7.1 Simplified Schematic of the S2000 System

The S2000 is linked to its ground-based winch station by a 2.5 km multipurpose conductive tether. This cable is a composite structure, featuring a high-tensile carbon-fibre core to withstand buoyancy and wind-drag forces, surrounding an aluminium-based conductive layer for power transmission. Onboard sensors relay real-time data through fibre-optic filaments within the tether, allowing a ground-based AI to adjust turbine speeds and aerostat pitch.

Table 7.1: S2000 SAWES System Specifications and Operating Parameters

Category	Parameter	Value (at h = 2000 m)
Atmospheric	Air density (ρ_{air})	1.007 kg m^{-3}
	Average Wind Speed (v)	15.0 m s^{-1}
Aerostat	External Diameter (D_{ext})	40 m
	Internal Diameter (D_{int})	30 m
	Volume of Helium (V_{He})	20,000 m^3
	Aerostat Structural Mass ($m_{structure}$)	5,000 kg
Generation	Total Turbine Swept Area ($A_{turbine}$)*	340 m^2
	Turbine Assembly Mass ($m_{turbine}$)*	2,400 kg
	Maximum Rated Power (P_{max})	3.0 MW
	Average Power Output from Initial Test	770 kW (over 30 min)
Tether	Total Length l_{tether}	2500 m
	Linear Mass	1.8 kg m^{-1}

*Note: Total of the 12 internal turbines.

Additional Constants and Parameters:

Aluminium resistivity: $\rho_{Al} = 2.8 \times 10^{-8}\ \Omega$ m

Helium density: $\rho_{He} =$ 0.18 kg m^{-3}

Power coefficient: $C_p = 0.40$ (maximum)

System efficiency: $\eta_{system} = 45\%$ (maximum)

Drag coefficient (ring aerostat): $C_d = 0.25$

Energy density of the stratospheric wind, which represents the kinetic energy per unit volume of moving air, helps engineers assess the energy resource available at different altitudes and locations. This, in turn, will help to determine the amount of power that can be harnessed from the wind through the turbine.

Real turbines cannot extract all available wind power. The power coefficient (C_p) represents the fraction of wind power actually converted to mechanical power. For example, a C_p of 0.40 means the turbine converts 40% of the wind's kinetic energy into mechanical rotation. System efficiency (η_{system}) accounts for subsequent losses during electrical conversion and transmission, reducing the final delivered power. This includes losses in the generator, power electronics, and the resistance of the 2.5 km tether. A system efficiency of 45% means only 45% of the mechanical power extracted by the turbines reaches the ground station as usable electricity.

(a) **(i)** Using data from Table 7.1, calculate the buoyant force (upthrust) $\rho_{air} V_{He} g$ acting on the helium-filled aerostat at its operating altitude.

buoyant force = ……. N [1]

(ii) Calculate the total weight of Helium-filled aerostat.

total weight = ……. N [2]

(iii) Using your answers from **(a)(i)** and **(a)(ii)**, determine the minimum tension in the tether.

minimum tension = ……. N [1]

(iv) Calculate the weight of the tether.

weight = ……. N [1]

(v) State whether the force exerted by the Winch on the tether will be greater, equal or smaller than your answer in **(a)(iii)**, explain your answer.

……..

……..

……..

……..… [2]

(b) **(i)** When wind flows around the ring-shaped aerostat, it experiences a drag force given by $\frac{1}{2}\rho v^2 C_d A$, where A is the frontal area presented to the wind. Assuming the frontal area is a circle with external diameter D_{ext}, calculate the drag force acting on the aerostat in the stratospheric wind conditions.

drag force = ……. N [2]

(ii) Determine the magnitude of the change in tension in the tether, when drag force acts on the aerostat. Explain your answer.

magnitude of change in tension = ……. N [2]

(c) **(i)** By considering the energy density of the stratospheric wind, show that the power that can be harnessed from the wind can be expressed as $\frac{1}{2}\rho_{air}A_{turbine}v^3$, where ρ_{air}, $A_{turbine}$ and v are quantities expressed in Table 7.1.

[2]

(ii) Determine the total power harnessed from the wind through the turbine.

total power = ……. W [1]

(iii) Hence or otherwise, determine maximum electrical power delivered to ground.

power = ……. W [2]

(iv) The average power output from the Initial test from Table 7.1 is 770 kW. State whether this is higher or lower than your calculated value in **(c)(iii)** and suggest a reason for the difference.

……..

……..

……..

……..

…… [2]

(d) Dimensions of the tether must be considered in order to carry a current of up to 300 A and heat losses through the tether only account for 0.48% of the maximum rated power $\left(P_{\text{max}}\right)$ for efficient transmission of power.

Determine the cross-sectional area of the aluminium needed in the conductive layer of the tether that satisfies the above considerations.

cross-sectional area = …….…… m^2 [2]

[Total: 20]

Section A

Answer **all** the questions in this Section in the spaces provided.

1 A battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance is connected in series with two fixed resistors and a thermistor, as shown in Fig. 1.1.

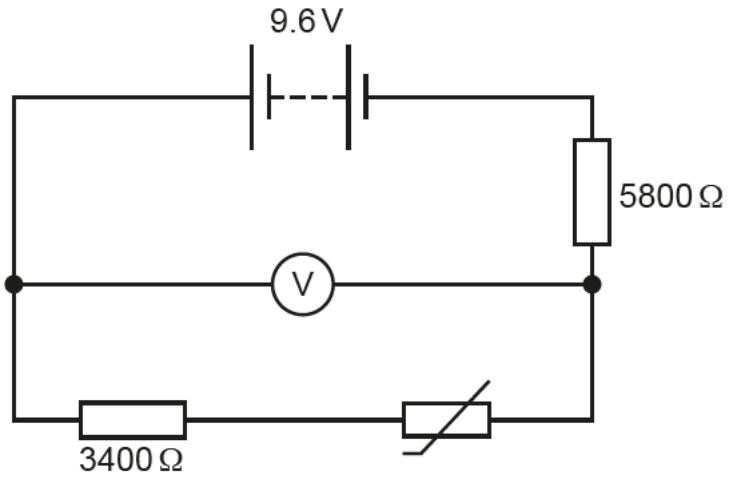


Fig. 1.1

The fixed resistors have resistances of 3400 Ω and 5800 Ω. The reading on the voltmeter in the circuit is 6.0 V.

(a) Calculate the current in the resistor of resistance 5800 Ω.

current = ... A [2]

(b) Calculate the resistance of the thermistor.

resistance = ... Ω [2]

(c) The initial energy stored in the battery is 2.6×10^4 J.

Assume that the e.m.f. of the battery is constant.

Determine the final energy stored in the battery after a charge of 330 C has moved through it.

final stored energy = ... J [2]

(d) The environmental conditions change causing an increase in the resistance of the thermistor.

State whether there is a decrease, increase or no change to:

(i) the temperature of the thermistor

…….. [1]

(ii) the current in the thermistor

…….. [1]

(iii) the potential difference across the thermistor.

…….. [1]

(e) If the temperature of the thermistor in Fig. 1.1 changes at a constant rate, state a reason why the potential difference across the thermistor does not change at a constant rate.

……...

……...

…… [1]

[Total: 10]

5 Two point charges, A and B, are separated in a vacuum by a distance of 0.0100 cm, as shown in Fig. 5.1.

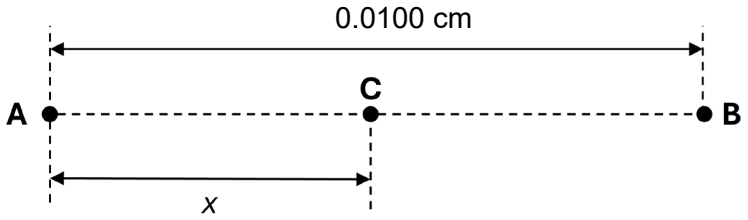


Fig. 5.1

A point C is on the line joining the two charges and is a distance *x* from charge A.

The variation with distance *x* of the electric potential V_x at point C is shown in Fig. 5.2.

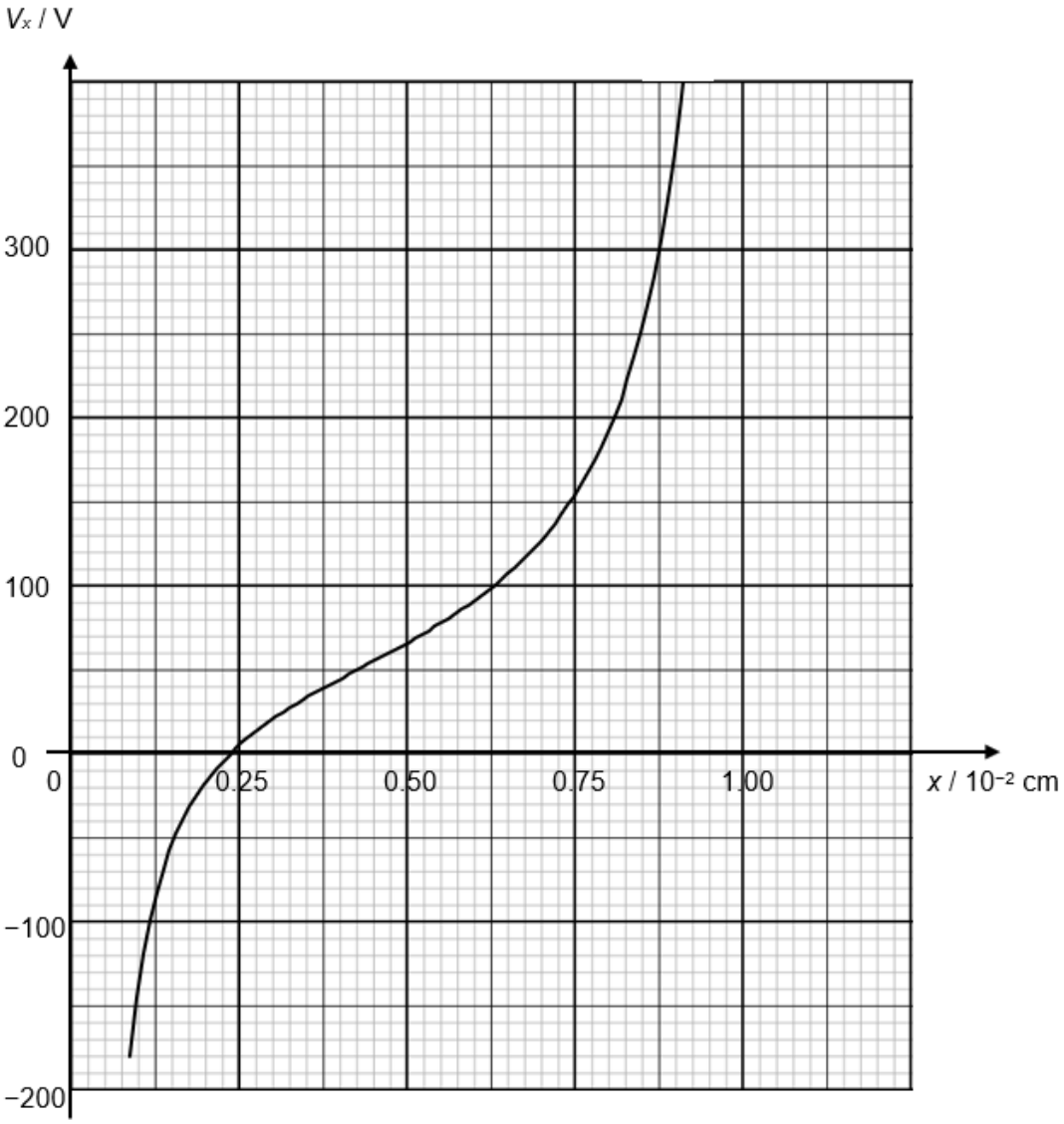


Fig. 5.2

(a) Define electric field strength at a point.

……..

……..

…… [1]

(b) Explain whether the charges have the same or opposite signs.

……..

……..

……..

……..

…… [2]

(c) Calculate the electric field strength where $x = 0.250 \times 10^{-2}$ cm.

electric field strength = ... V m^{-1}

direction of electric field strength: ..... [3]

(d) An electron is projected with a speed of 7.73×10^{6} m s^{-1} from the position where $x = 0.725 \times 10^{-2}$ cm towards point charge A.

Calculate the speed of the electron at the position where $x = 0.175 \times 10^{-2}$ cm.

speed = ... m s^{-1} [3]

[Total: 9]

6 A simple ideal iron-cored transformer is illustrated in Fig. 6.1.

An alternating input p.d. V_p of 240 V is connected across the primary coil, of 120 turns of the transformer. The secondary p.d. V_s of 3600 V is applied to a load resistor.

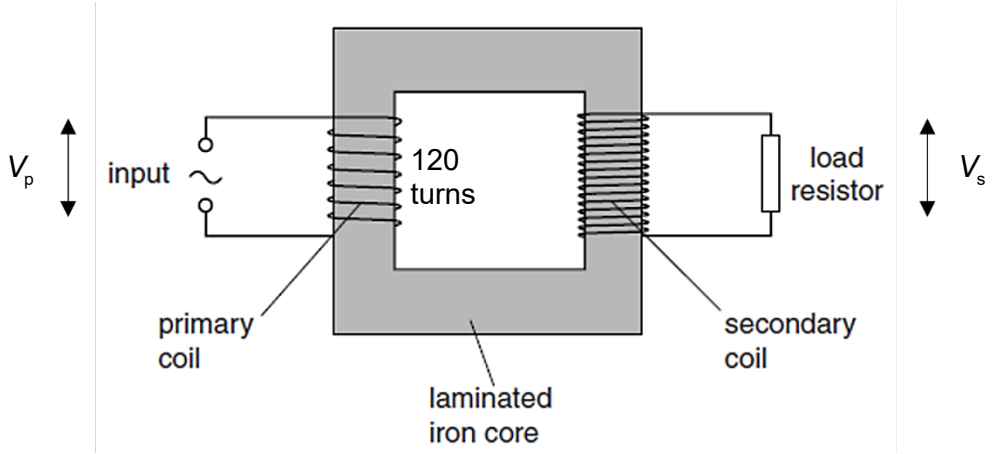


Fig. 6.1

(a) **(i)** Explain why an iron core is preferred over an air core in the construction of a transformer.

.....

.....

..... [1]

(ii) Explain why the iron core is laminated.

.....

.....

..... [1]

(b) Calculate the number of turns in the secondary coil.

number of turns = ... [1]

(c) Using Faraday's Law, explain why an alternating current in the primary coil gives rise to an alternating e.m.f. in the secondary coil.

……..

……..

……..

……..

……..

……..

……..

……..

……..

……..

…… [3]

(d) A step-up transformer is used when transmitting electricity over long distances.

Suggest a reason for this.

……..

……..

……..

……..

……..

……..

…… [1]

[Total: 7]

3 **(a)** Define *electric potential* at a point in an electric field.

……

……..…[1]

(b) Two charged spheres of radius 2.0 mm are held 4.00 cm apart (centre-to-centre). Fig. 3.1 shows the variation with displacement x, of the electric potential V, between the two charged spheres. The charge on each sphere may be considered to be concentrated at its centre.

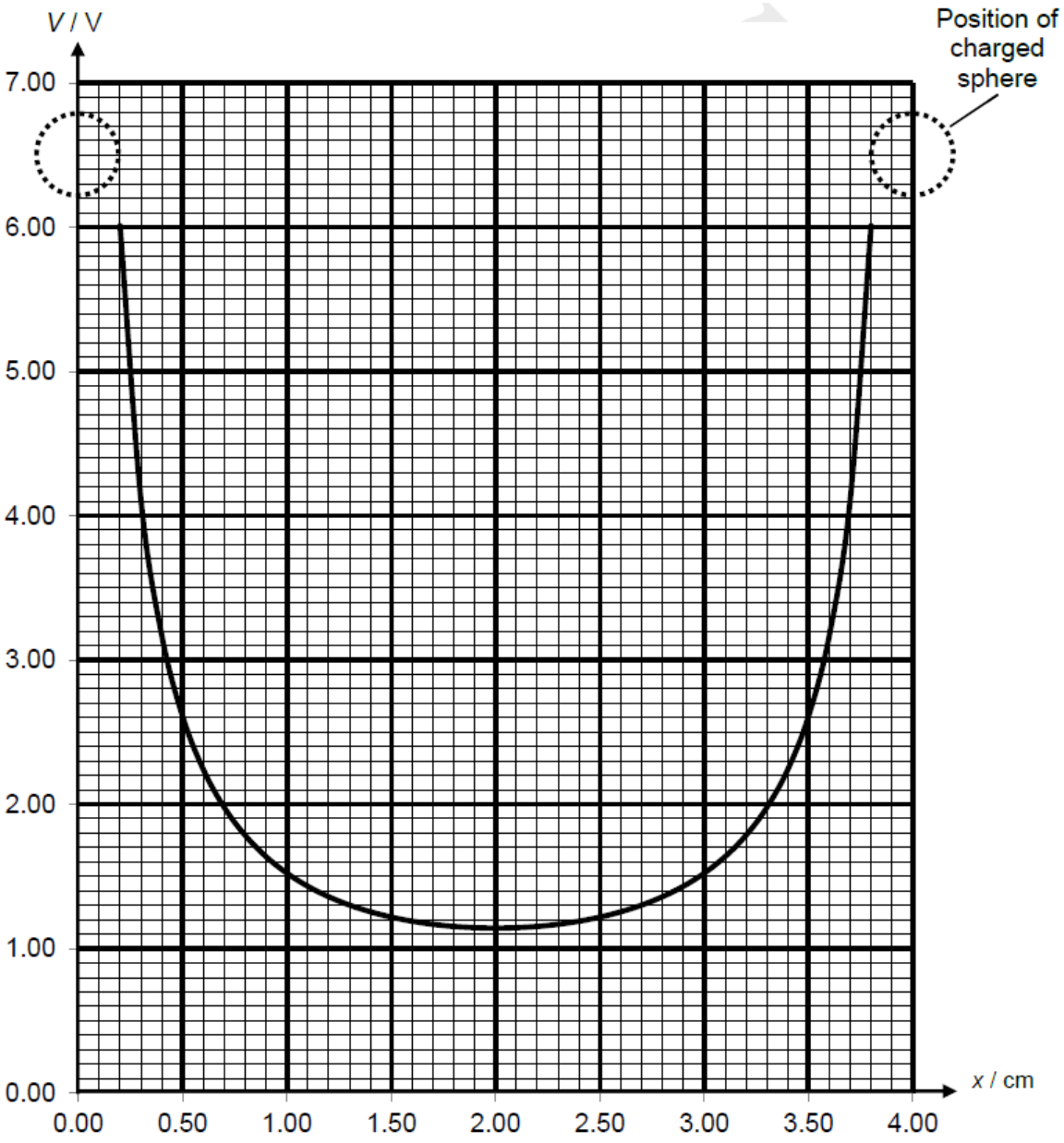


Fig. 3.1

(i) Using Fig. 3.1, determine the magnitude of the electric field strength at x = 1.00 cm and state its direction.

electric field strength = …… N C^{-1} [2]

direction: …… [1]

(ii) Explain, using Fig. 3.1, how it can be deduced that the charges on the two spheres are equal.

……

……[1]

(iii) Using Fig. 3.1, determine the charge on each sphere.

charge = …….. C [2]

[Total: 7]

4 **(a)** Define the resistance of a circuit component.

……

…… [1]

(b) With reference to energy transfers, distinguish between electromotive force (e.m.f.) and potential difference (p.d).

……

……

…… [1]

(c) Fig. 4.1 shows a circuit containing cell of e.m.f. E with internal resistance r, a fixed resistor with resistance R_1 and a variable resistor with resistance R_2.

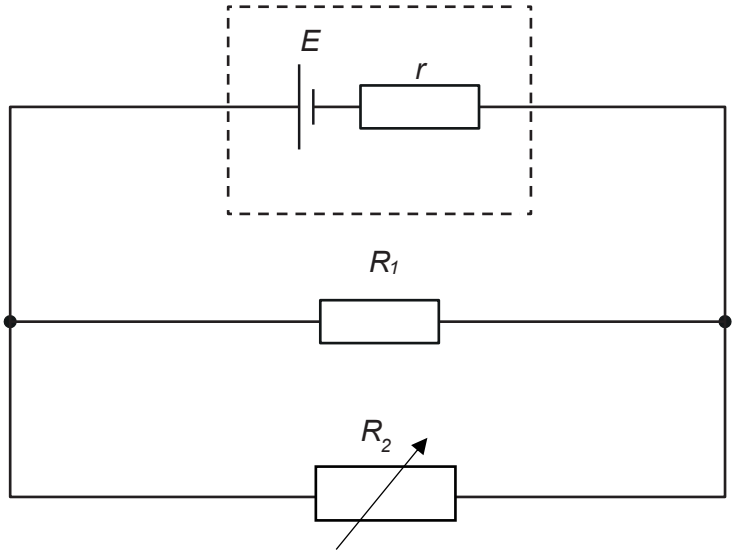


Fig. 4.1

With reference to the current in the cell, explain what happens to the current through R_1 when R_2 increases.

……

……

……

……

……

…… [3]

(d) The e.m.f. E of the cell in Fig. 4.1 is 1.40 V. The internal resistance r of the cell is 0.13 Ω.

R_1 has a resistance of 5.50 Ω.

At a particular resistance R_2, the current through R_1 is measured to be 0.220 A.

Determine

(i) the current through the cell

current = ... A [2]

(ii) the resistance R_2.

R_2 = ... Ω [2]

[Total: 9]

6 **(a)** State Faraday's law of electromagnetic induction.

……

……

…… [1]

(b) An aircraft is flying horizontally at constant speed *v* through the Earth's magnetic field, as shown in Fig. 6.1.

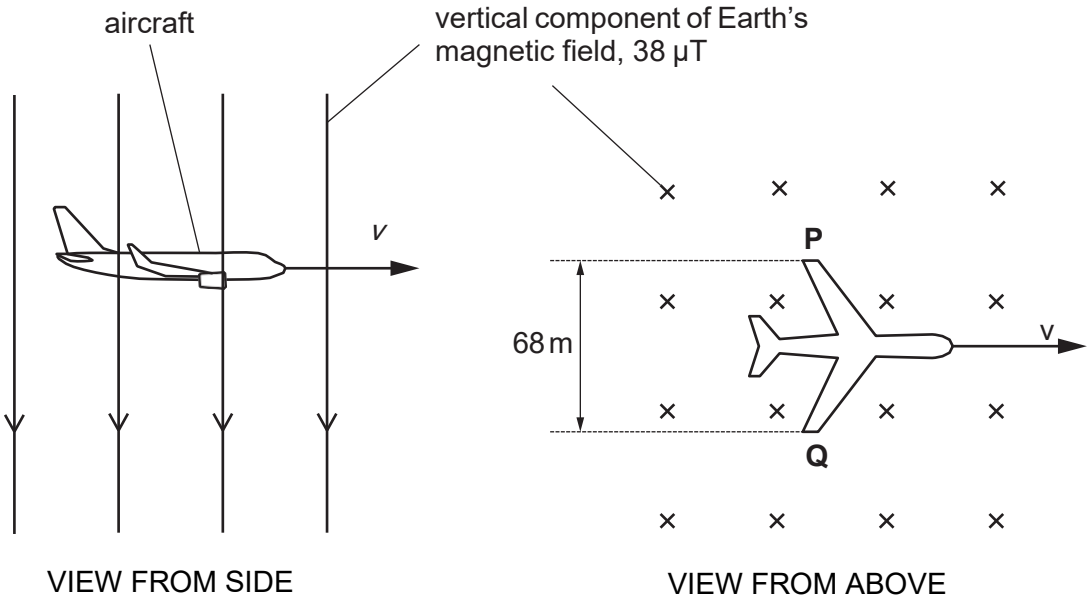


Fig. 6.1

At the location of the aircraft, the vertical component of the Earth's magnetic field is 38 μT towards the ground.

The distance between the wingtips P and Q of the aircraft is 68 m.

As the aircraft moves through the magnetic field, an electromotive force (e.m.f.) of 0.55 V is induced between the wingtips P and Q.

(i) Calculate the magnetic flux cut by the wings of the aircraft in a time of 14 s.

Provide a suitable unit with your answer.

magnetic flux = ……

unit = …….. [2]

(ii) Determine the area of flux cut by the wings in a time of 14 s.

area = …… m^2 [1]

(iii) Use your answer in **(b)(ii)** or otherwise to determine the speed v of the aircraft.

v = …… $m\ s^{-1}$ [2]

(iv) Using Fleming's Left Hand Rule, explain clearly why the wingtip P is at the higher potential compared to wingtip Q.

……

……

……

……

…… [2]

(c) A sinusoidal power source with alternating voltage V varies with time t according to

$$V = 16\cos(38\pi t)$$

where V is in V and t is in s.

The circuit connection with the two resistors R_1 and R_2 each having resistance R and an ideal diode is shown in Fig. 6.2

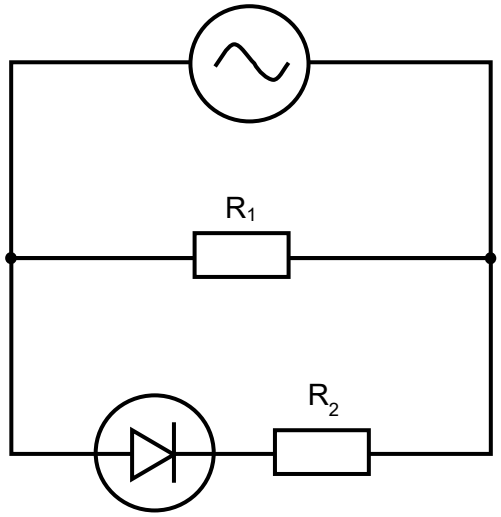


Fig. 6.2

(i) Fill in the missing values in Table 6.1.

resistor	r.m.s. voltage, $V_{r.m.s.}$ / V	peak voltage, V_0 / V
R_1		
R_2		

Table 6.1

[1]

(ii) Compare the r.m.s voltages and peak voltages across resistors R_1 and R_2, and explain your answers.

……

……

……

……

……

…… [2]

[Total: 11]

8 Read the passage and answer the questions that follow.

The human heart beats in response to electrical impulses generated by specialised cells in the heart itself. In a healthy heart, these impulses travel through a natural pathway that coordinates each contraction, driving the heart to beat between 60 and 100 times per minute at rest, as shown in Fig. 8.1 below.

In a condition known as complete heart block, this pathway is disrupted. The lower chambers no longer receive reliable electrical signals, causing the heart to beat slowly and erratically, sometimes fewer than 40 times per minute. This is insufficient to sustain blood circulation, and the patient faces the risk of sudden cardiac arrest.

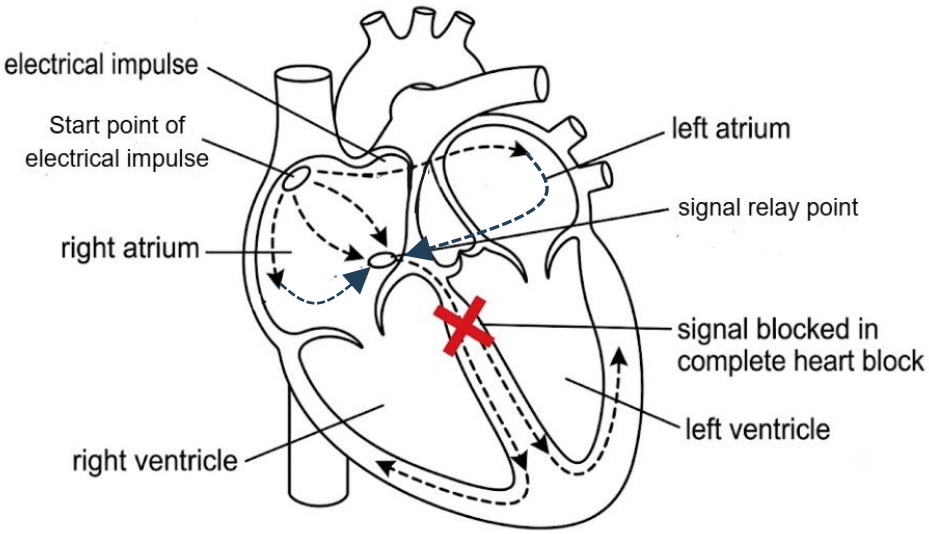


Fig. 8.1

In 1960, electrical engineer Wilson Greatbatch and surgeon William Chardack implanted the world's first fully implantable pacemaker into a human patient.

The device delivered regular electrical pulses to one of the heart's lower pumping chambers (the ventricles) via electrode leads, bypassing the blocked pathway and restoring a steady heart rate of 60 beats per minute. The timing of these pulses was controlled by a simple RC circuit known as a relaxation oscillator.

The pacemaker's timing circuit (RC Relaxation Oscillator) consists of a battery of electromotive force (e.m.f.) ε and a resistor R connected in series, with a capacitor C connected across the output. A voltage-controlled switch S is connected to switch between charging and discharging of the capacitor C.

A simplified circuit diagram is shown in Fig. 8.2 below:

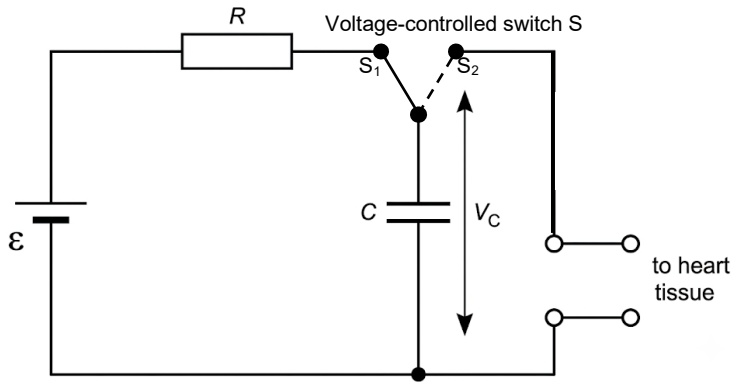


Fig. 8.2

The voltage-controlled switch S responds to the potential difference V_c across the capacitor. It is triggered to S_2 when V_c rises to an upper threshold voltage V_b, and switches to S_1 again when V_c falls to a lower threshold voltage V_s, where $V_s < V_b < \varepsilon$.

When switch S is at S_1, the battery charges the capacitor through the resistor.

When V_c rises to V_b, the switch triggers to S_2 almost instantaneously, and the capacitor discharges rapidly through the heart tissue. This brief discharge delivers a voltage pulse to the heart tissue via output electrodes.

During this discharge, when V_c falls to V_s, the switch returns to S_1 and the capacitor begins recharging. The cycle then repeats, generating a regular train of pulses.

The resulting variation of V_c with time t is shown in Fig. 8.3.

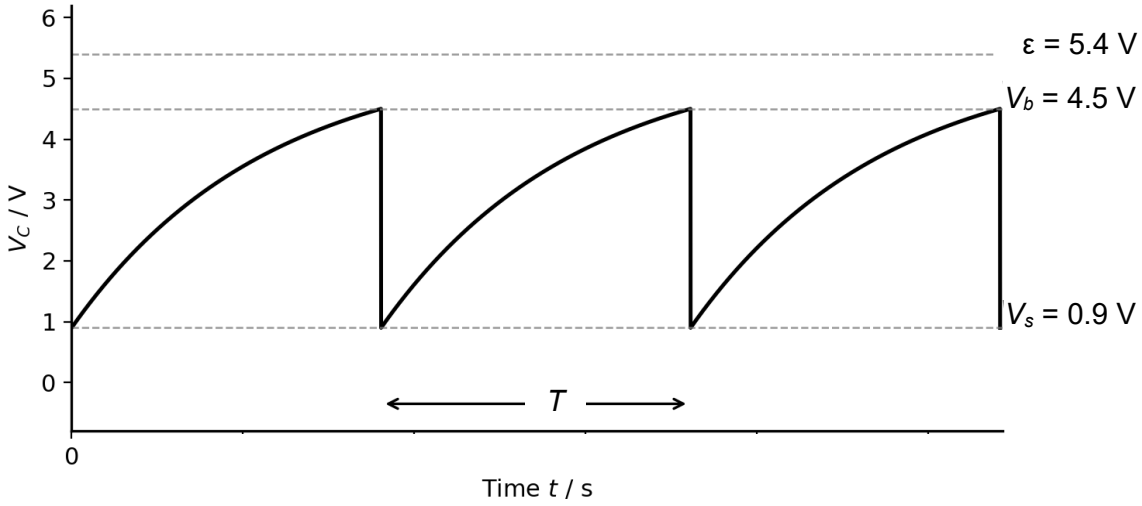


Fig. 8.3

Because the discharge time is very short compared to the charging time, the period T of the oscillation is approximately equal to the time taken for V_c to charge from V_s to V_b. It can be shown that:

$$T \approx RC \ln\left(\frac{\varepsilon - V_S}{\varepsilon - V_b}\right)$$

(a) **(i)** The component values used in the Greatbatch pacemaker are given in the table below.

Component	**Value**
Timing resistor R	18 kΩ
Timing capacitor C	33 μF

Hence, determine the pacing frequency of the Greatbatch pacemaker and comment whether this value is appropriate for a resting patient. State one limitation of this fixed-rate pacemaker.

Comment: ……

……....

Limitation: ……

……[4]

(ii) The resistance of the heart tissue is approximately 500 Ω. By calculating the charging and discharging time constants τ, comment on whether the assumption that the discharge is almost instantaneous is justified.

……

……[2]

(b) Each time the switch closes, the capacitor discharges from V_b to V_s, delivering a pulse of energy to the heart tissue.

(i) Show that the energy delivered per pulse is 3.21×10^{-4} J. [2]

(ii) Hence, using your answer to (a)(i) and (b)(i), calculate average power delivered to the heart tissue in one charging-discharging cycle.

average power = …… W [2]

(iii) The average power delivered during each discharge pulse is approximately 12.1 mW.

Explain why the average power delivered to the heart tissue in a charging-discharging cycle is much lower than the average power delivered during a discharge pulse.

……

……

……

……[2]

(c) The capacity of a battery is the total charge it can deliver, expressed in ampere-hours (Ah), where 1 Ah = 3600 C.

The original mercury-zinc battery used in this pacemaker had a capacity of 1.2 Ah at an e.m.f. of 5.4 V.

(i) Calculate the total electrical energy stored in the mercury-zinc battery.

energy = ……. J [2]

(ii) Estimate the expected operational lifespan of the battery in years. Comment on your result.

……

……

……[2]

(d) Over many years of operation, the capacitance of the timing capacitor will gradually increase due to physical degradation of its dielectric material.

(i) Explain qualitatively why that would result in a decrease in pacing frequency.

……

……

……

……[1]

(ii) Suggest, with reasoning, whether a slow decrease in pacing frequency over time would be dangerous for a patient with complete heart block.

……

……

……

……[2]

[Total: 19]

SECTION B

Answer any ONE of the two questions

8 A particle of mass m and charge q, is initially at rest at the midpoint between two parallel plates.

A potential difference V_0 is applied across the plates.

The particle accelerates towards the upper plate and enters into a region with a uniform magnetic flux density B, directed perpendicularly out of the page. There is no magnetic field anywhere else. A snippet of the particle's path is shown in Fig. 8.1.

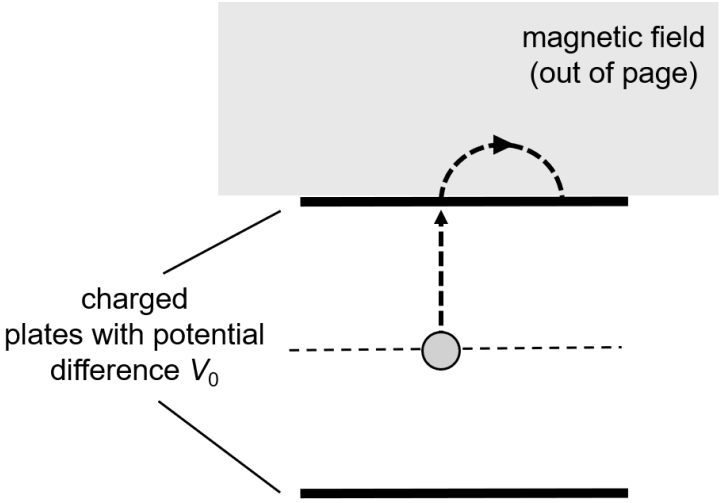


Fig. 8.1

(a) Explain

1. why the charged particle in Fig. 8.1 is positively charged,

……

……

……

…… [2]

2. and which plate has a higher electric potential.

……

……

……

…… [2]

(b) **(i)** Show that the magnitude of the potential difference between the top plate and the midpoint between the plates is:

$$\Delta V = \frac{V_0}{2}$$

[1]

(ii) Using your answer in **(b)(i)**, show that the velocity of the particle when it first enters the magnetic field is:

$$v = \sqrt{\frac{qV_0}{m}}$$

[1]

(c) Explain whether the speed of the particle increases while it is in the magnetic field.

……

……

…….. [2]

(d) Show that the time travelled in the magnetic field can be expressed as

$$t = \frac{m\pi}{Bq}$$

[4]

(e) The mechanism shown in Fig. 8.1 forms the basis of a classical cyclotron.

Instead of two parallel plates, the cyclotron consists of two hollow, D-shaped metal electrodes (known as dees) separated by a narrow central gap. The dees are housed in a vacuum chamber within the uniform magnetic field of flux density *B* directed perpendicularly out of the page.

An alternating high-voltage source produces an electric field across the gap.

When the charged particle is released at the centre, it is accelerated across the gap into the upper dee. Shielded from the electric field inside the hollow metal dee, the magnetic field bends the particle into a semicircular path.

Upon exiting back into the gap, the electric field reverses polarity, accelerating the particle into the lower dee at a higher speed. This process repeats, producing an expanding spiral path until the particle exits the cyclotron.

This is shown in Fig. 8.2.

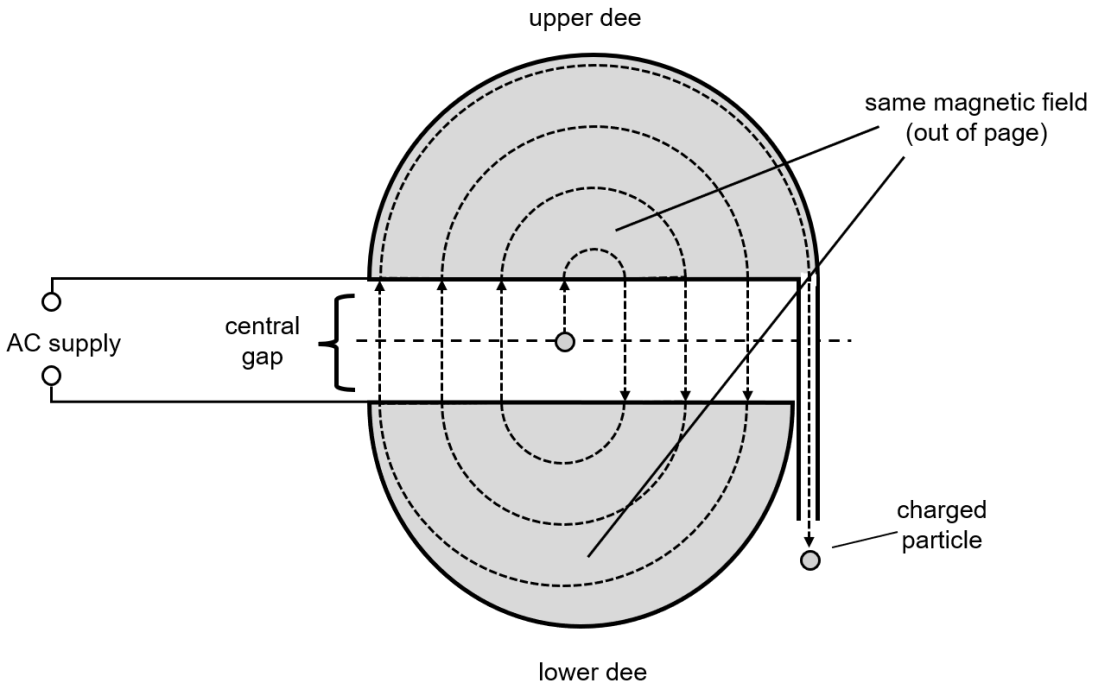


Fig. 8.2

A man claims that the time travelled in each dee is the same as that found in **(d)**.

State and explain if his claim is true.

……

……

……

…….. [2]

(f) The alternating voltage supply is a square wave that switches its polarity at constant time intervals equal to the time taken for the particle to complete a semicircle inside each dee.

This is shown in Fig. 8.3.

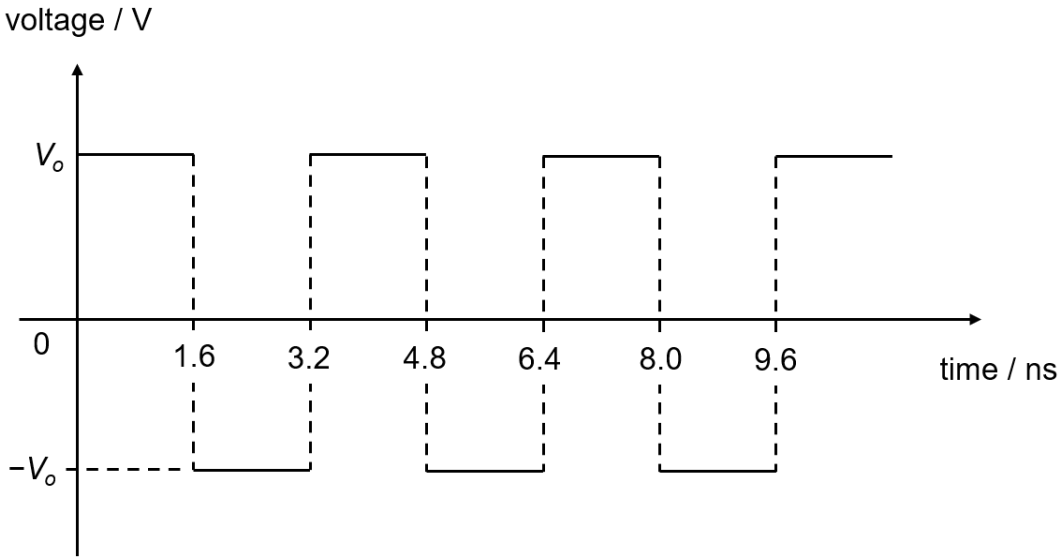


Fig. 8.3

Given that the charge to mass ratio of the particle is 1.0×10^8 C kg^{-1}, using the equation in **(d)**, determine the magnetic flux density in the dees.

You may assume that the time taken for the charged particle to travel the central gap is negligible.

magnetic flux density = ... T [2]

(g) Determine the kinetic energy of the charge when it first enters the lower dee, leaving your answer in q and V_0.

kinetic energy = ... [2]

(h) The particle leaves the cyclotron with a kinetic energy of 6.5 qV_0.

Determine the number of times it travels in the lower dee before leaving the cyclotron.

number = ... [2]

[Total: 20]

5 A small conducting sphere of mass of 0.12 g is suspended from a very long insulating thread between metal plates M and N, separated by 2.0 cm as shown in Figure 5.1.

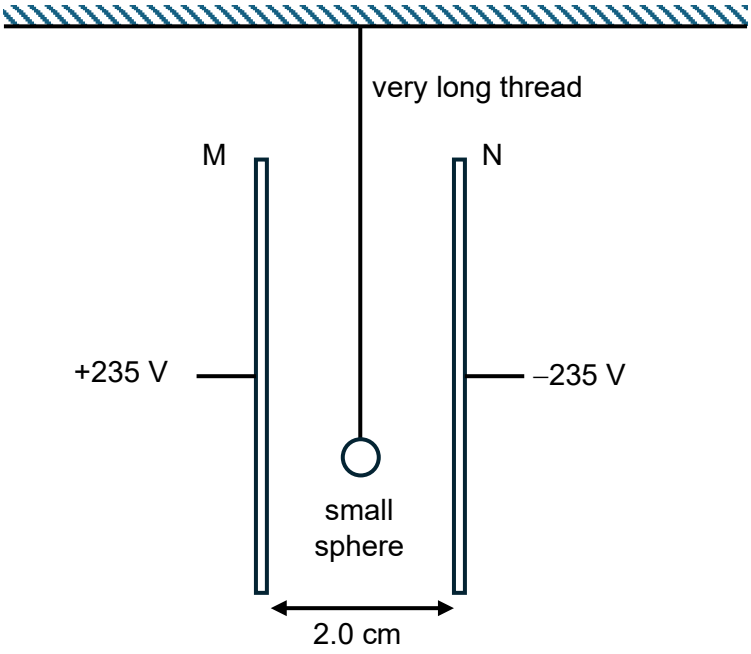


Figure 5.1 (not to scale)

Plates M and N are connected to a high-voltage power supply and are held at potentials of 235 V and -235 V, respectively.

After being given an initial displacement, the sphere moves to plate M, touches it and acquires a charge of 58 pC.

It then moves towards plate N, touches it and the horizontal oscillation process repeats itself.

Assume that the initial velocity of the sphere is negligible when moving away from the plates.

(a) Calculate the electrostatic force on the sphere as it moves towards plate N.

force = …….. N [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) Calculate the acceleration of the sphere.

acceleration = ... m s^{-2} [1]

(c) Determine the frequency of oscillation of the sphere. Assume the size of the sphere is negligible compared to the separation of the plates.

frequency = ... Hz [3]

(d) A student claims that the oscillation is simple harmonic.

Explain whether the student is correct.

.....

.....

.....

..... [1]

[Total: 7]

6 Figure 6.1 shows a circuit that produces rectification of an alternating input voltage.

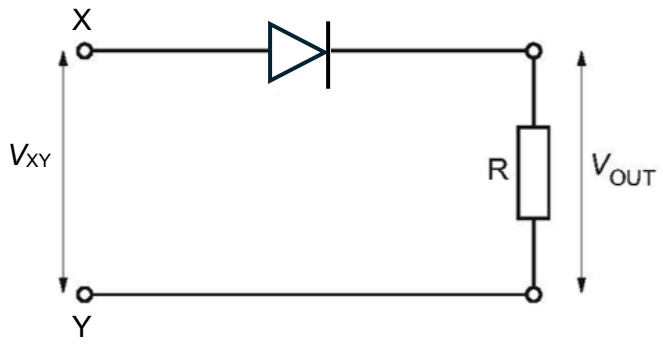


Figure 6.1

(a) The input voltage V_{XY} is the potential of X with respect to Y and is sinusoidal. V_{OUT} is the rectified voltage across resistor R.

The variation of V_{XY} with time t has amplitude V_0 and period T, as shown in Figure 6.2.

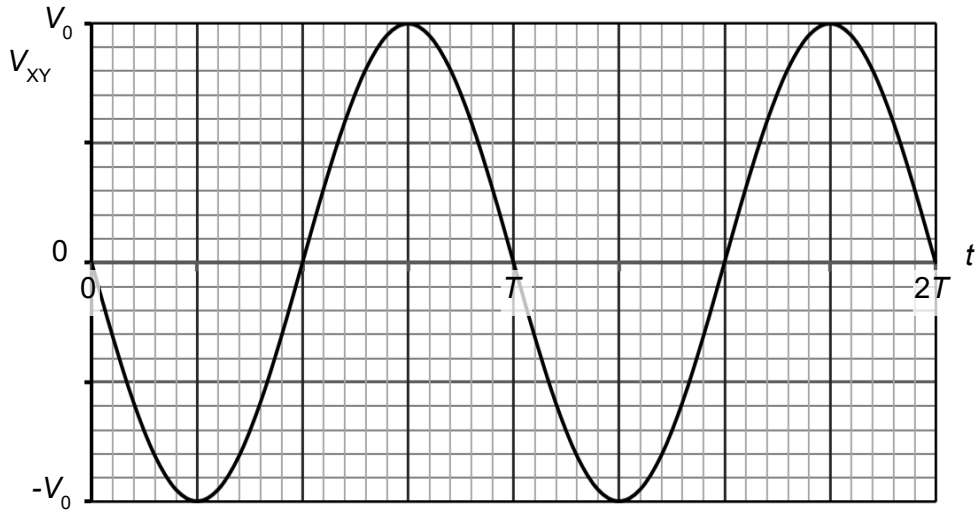


Figure 6.2

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(i) On Figure 6.3, sketch the variation with time t of the power P in the resistor between $t = 0$ and $t = 2T$.

Label the peak power P_0 in the resistor on your sketched graph.

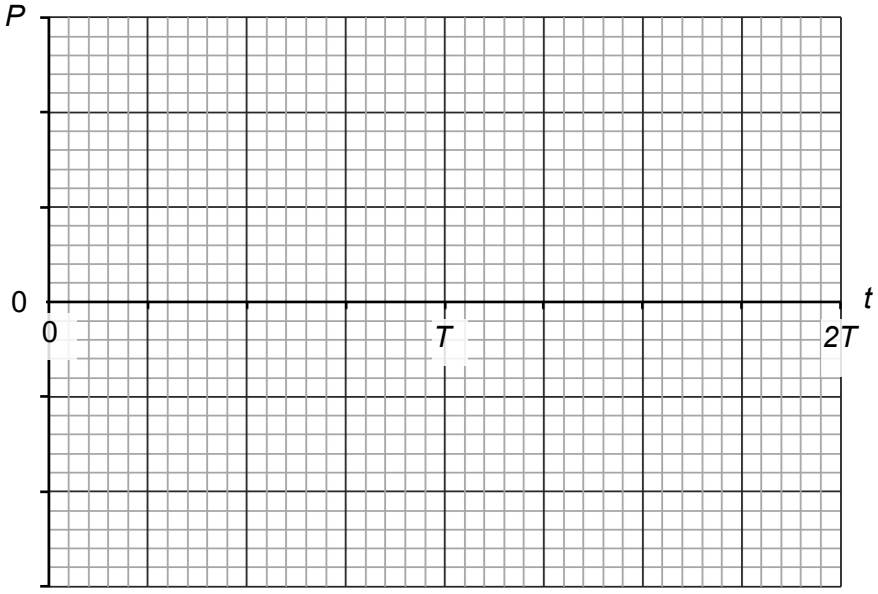


Figure 6.3

[2]

(ii) Use the answer in **(a)(i)** to explain why the mean power in the resistor is $\frac{1}{4}P_0$.

.....

.....

.....

..... [2]

(iii) The root-mean-square (r.m.s.) value of V_{XY} is 12 V.

Use the information in **(a)(ii)** to show that the r.m.s. value of V_{OUT} is 8.5 V.

[2]

(b) To smooth the output voltage, a capacitor is connected in parallel to the resistor in Figure 6.1.

During each discharge cycle, the minimum value of the smoothed output voltage is 0.60 V_0,

(i) Figure 6.4 shows a duplicate of the graph of V_{XY} with time t.

On Figure 6.4, sketch the variation with time t of the smoothed rectified output voltage V_R across the resistor R between $t = 0$ and $t = 2T$.

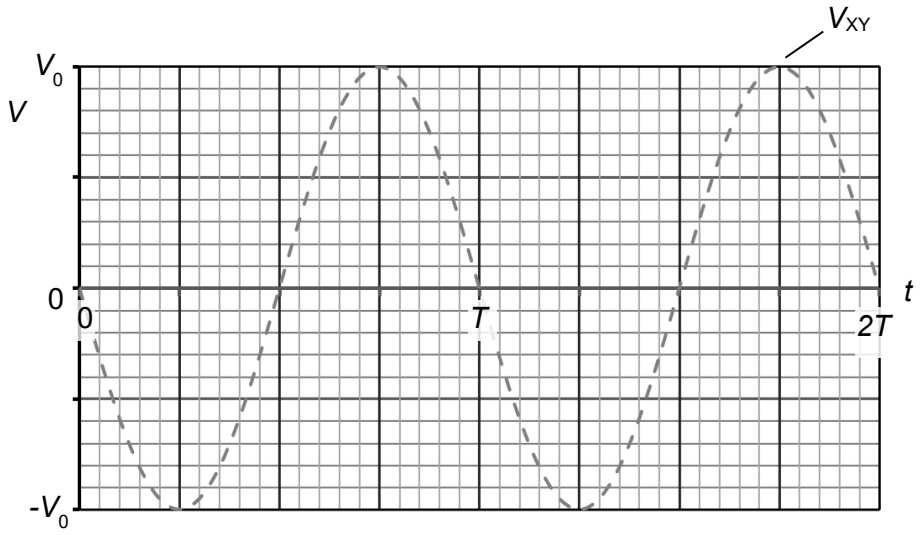


Figure 6.4

[1]

(ii) Determine the time constant of the smoothing circuit in terms of the period T.

time constant = ... [2]

[Total: 9]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

5 Student X constructs a circuit to demonstrate the effect of loading a battery by adding resistors in parallel across it. An ammeter of negligible resistance measures the total current drawn from the battery.

Resistors can be added into the circuit by closing switches **A**, **B**, **C**, **D**, **E** as shown in Figure 5.1.

The battery has an e.m.f. of 9.0 V and each of the resistor has a labelled value of 10 Ω.

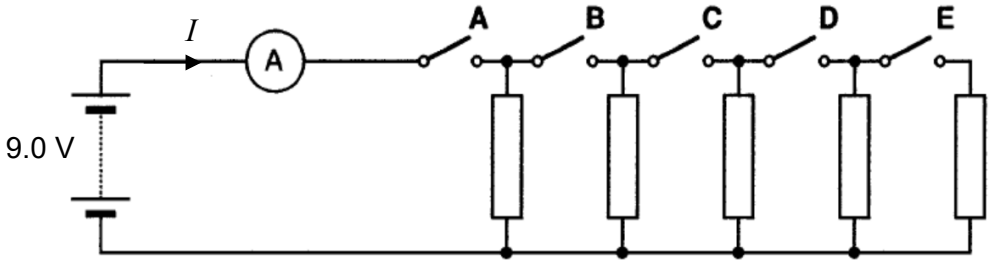


Figure 5.1

(a) Due to the way the resistors are made, there is a random variation of ± 5% in their value.

Student X assumes that the internal resistance of the battery is negligible.

Using this assumption and taking the e.m.f. of the battery to be constant, show that the possible variation in the value of the current I when only switch **A** is closed is approximately 0.05 A.

[2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) Figure 5.2 shows Student X's predicted and actual results.

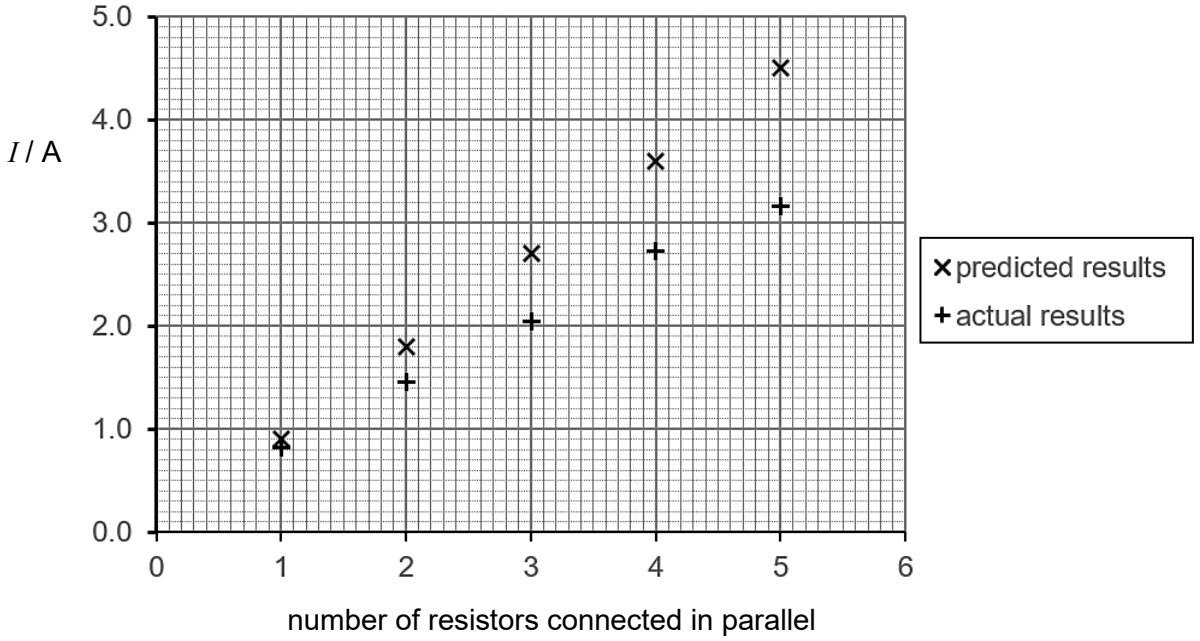


Figure 5.2

(i) Student X suggests that the difference between the two sets of results is due to the random variation (± 5%) of the value of the resistors.

Explain how the data in Figure 5.2 shows that Student X is **incorrect**.

.....

.....

..... [1]

(ii) Student Y then suggests that the increasing deviation between predicted and actual results are due to the internal resistance of the 9.0 V battery.

Explain how the data in Figure 5.2 supports Student Y's statement.

.....

.....

.....

.....

.....

..... [2]

(c) Student Y then plots a graph of $1/I$ against the resistance R of the resistors in parallel to find the internal resistance of the battery. (Figure 5.3)

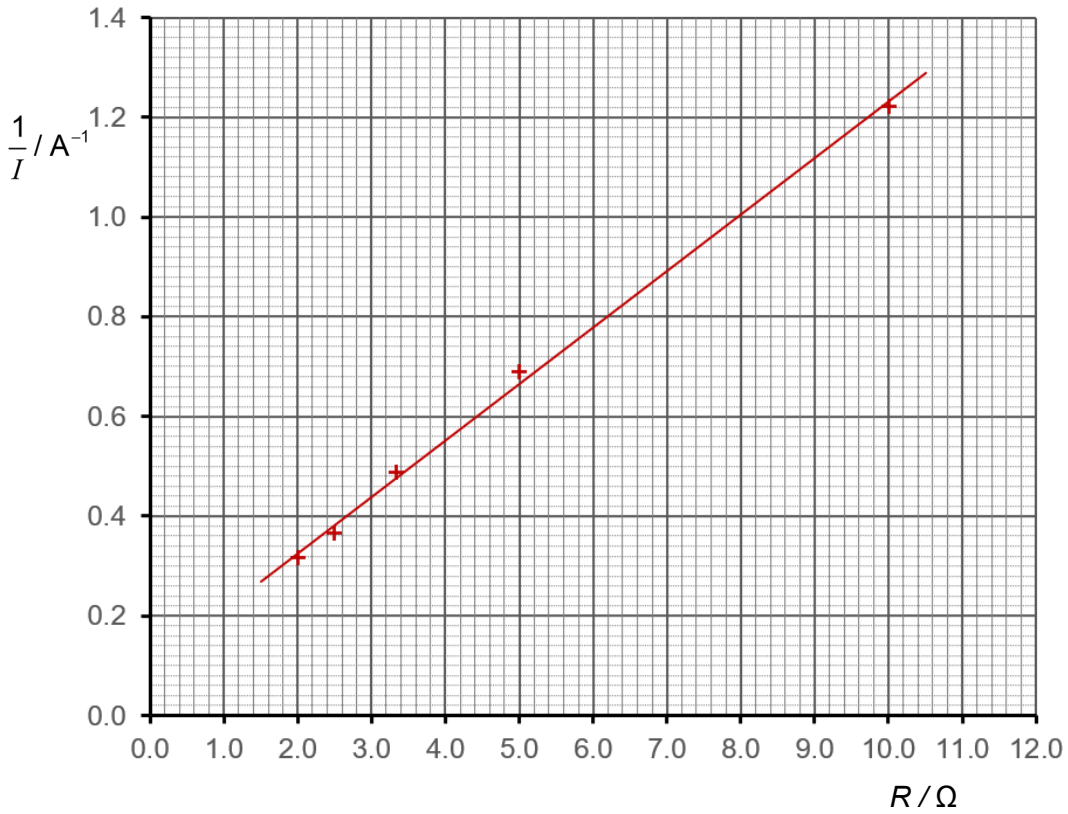


Figure 5.3

Use the graph to determine the internal resistance of the battery.

internal resistance = ... Ω [3]

[Total: 8]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

6 A solenoid is connected to an e.m.f source and a variable resistor as shown in Figure 6.1. A small coil connected to a galvanometer is placed close to the solenoid.

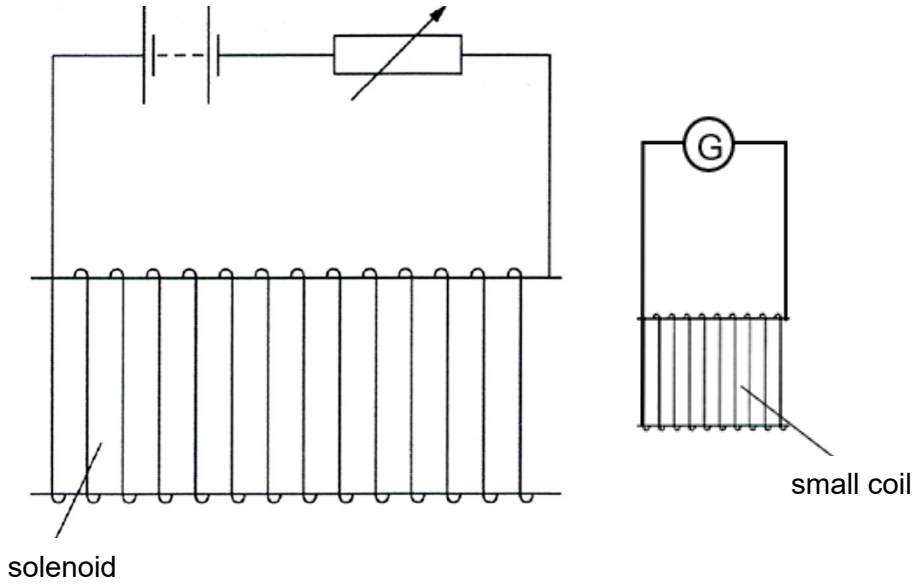


Figure 6.1

(a) The resistance of the variable resistor is slowly increased.

(i) State Faraday's law of electromagnetic induction and use it to explain why a reading is detected in the galvanometer.

.....

.....

.....

..... [2]

(ii) On Figure 6.1, mark the direction of the induced current passing through the small coil.

Explain the direction of the induced current.

.....

.....

.....

.....

..... [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) The current in the solenoid is now made to vary as shown in Figure 6.2.

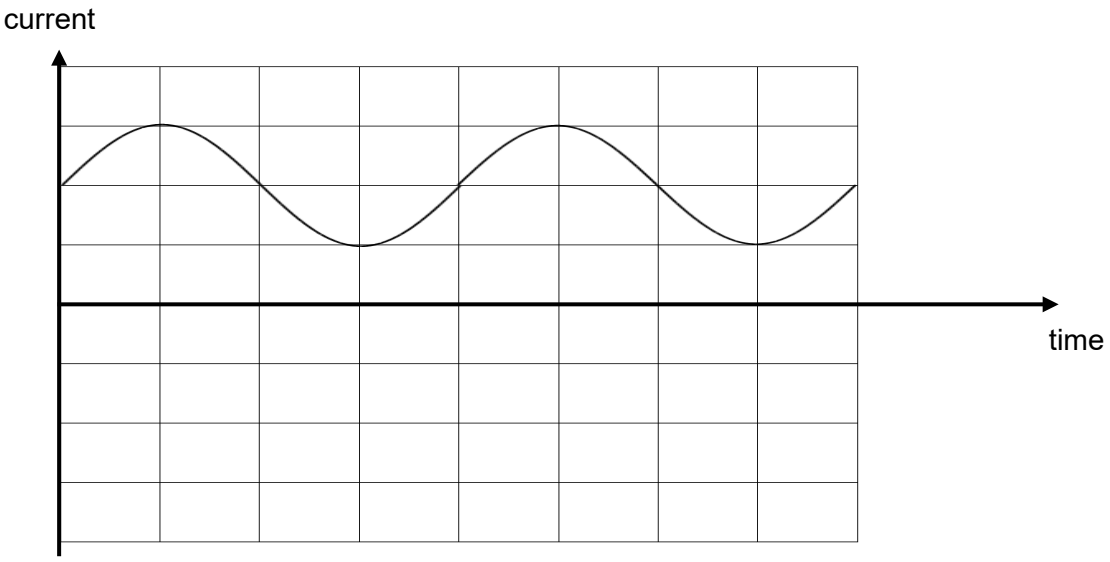


Figure 6.2

(i) On Figure 6.3, sketch the variation with time of the e.m.f. induced in the small coil. Label it as **P**. The time scale is the same. [1]

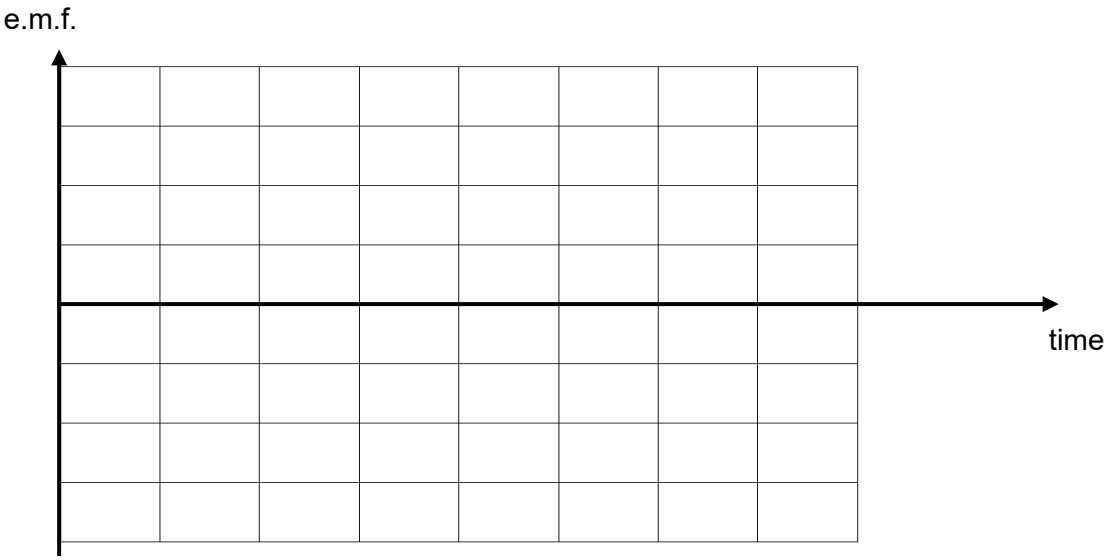


Figure 6.3

(ii) The frequency of the current in the solenoid is halved.

On Figure 6.3, sketch the new variation of the induced e.m.f. in the small coil and label it as **Q**.

[2]

[Total: 7]

4 **(a)** Fig. 4.1 shows a circular coil of 500 turns and radius 0.12 m.

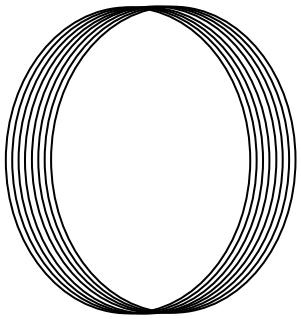


Fig. 4.1

A varying magnetic field is applied at right angles to the plane of the coil. Fig. 4.2 shows the variation with time *t* of this magnetic field of flux density *B*.

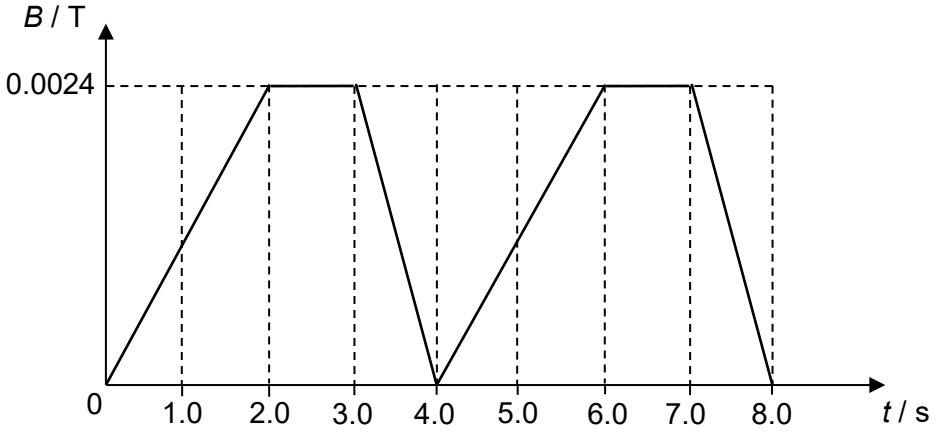


Fig. 4.2

(i) On Fig. 4.3, sketch a graph to show the variation with time t of the induced e.m.f. *E* in the coil. Include values on the y axis.

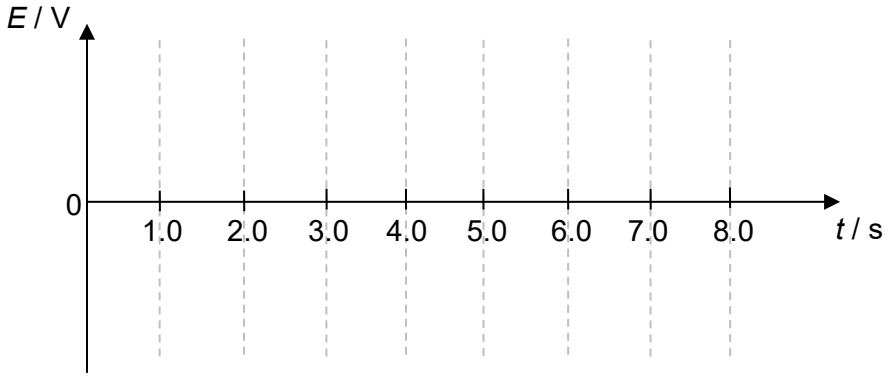


Fig. 4.3

[3]

(ii) Hence, determine the r.m.s value of the induced e.m.f.

r.m.s. value of induced e.m.f. = ... V [2]

(iii) The e.m.f. source that you have sketched in Fig. 4.3 is now connected in series with a resistor *R* and an ideal diode as shown in Fig. 4.4.

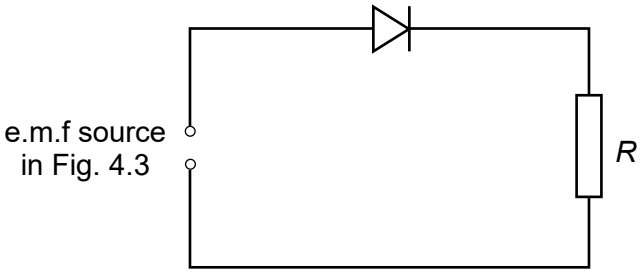


Fig. 4.4

State and explain the potential difference across R when the ideal diode has maximum potential difference across it.

.....

.....

.....

.....[3]

[Total: 8]

5 **(a)** A straight rigid conducting wire of mass 0.14 kg and length 0.40 m is placed horizontally on a frictionless plastic plane. The plane is at an angle of 20° to the horizontal and is placed in a region of upwards uniform vertical magnetic field as shown in Fig. 5.1 and Fig. 5.2.

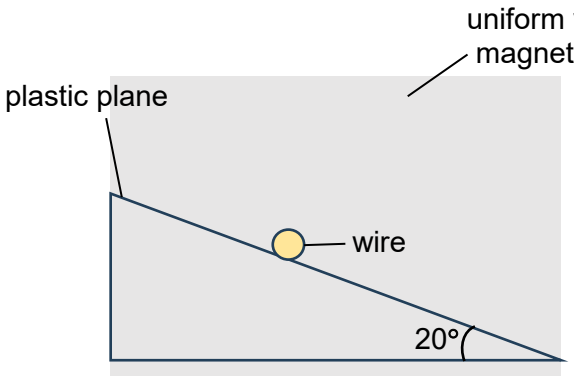


Fig. 5.1 (side view)

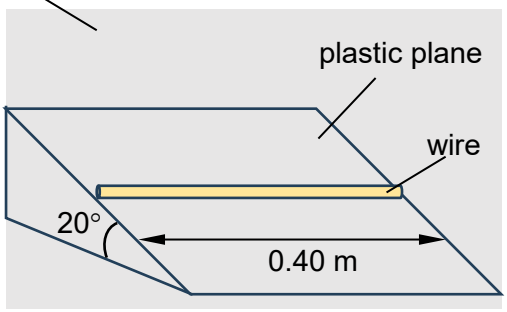


Fig. 5.2

To prevent the wire from sliding down the plane, a steady current I is passed through the wire so that it remains stationary on the plastic plane.

(i) On Fig. 5.1, draw an arrow to show the direction of the magnetic force acting on the wire. [1]

(ii) On Fig. 5.2, draw an arrow to show the direction of the current I in the wire. [1]

(iii) With reference to forces acting on the wire, explain why the wire remains stationary.

.....

.....

.....

.....

.....[2]

(iv) Suggest and explain whether the wire will eventually slide up, remain stationary or slide down the plane if an alternating current of root-mean-square value of I is passed through the wire instead.

.....

.....

.....

.....

.....[2]

(b) The current in **(a)** is disconnected, and the magnetic field is removed. The wire is now given a uniform net positive charge of $+2.0 \times 10^{-6}$ C.

(i) To keep the wire stationary in the same position on the frictionless plane, a uniform electric field of strength E is applied in the horizontal direction.

Determine the magnitude of E.

E = ... N C^{-1} [3]

[Total: 9]

9 **(a)** In a circuit, a 4.0 V battery with internal resistance is connected in series to a thermistor and a coaxial resistor.

The coaxial resistor has a cylindrical core of length 0.020 m. The core is made of material A and is surrounded by material B as shown in Fig. 9.1. In the circuit, the direction of the current passing through the coaxial resistor is parallel to its length. The physical dimensions of the coaxial resistor are fixed.

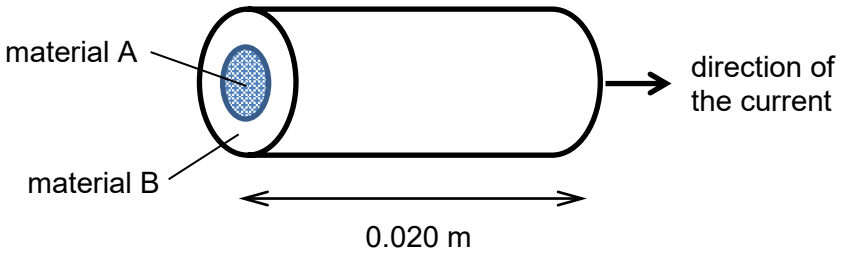


Fig. 9.1

(i) Draw a complete circuit diagram of the circuit in the space below.

Represent the coaxial resistor as two separate resistors A and B, one for each material respectively.

[3]

(ii) Fig. 9.2 shows the variation of the current I in the thermistor with temperature T of the thermistor.

Fig 9.3 shows the variation of I in the thermistor with voltage V across the thermistor.

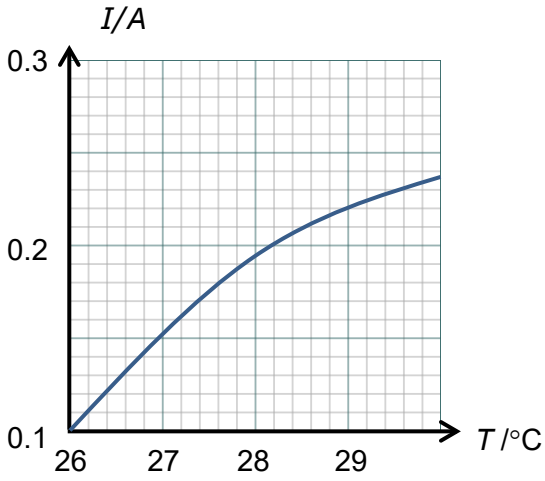


Fig. 9.2

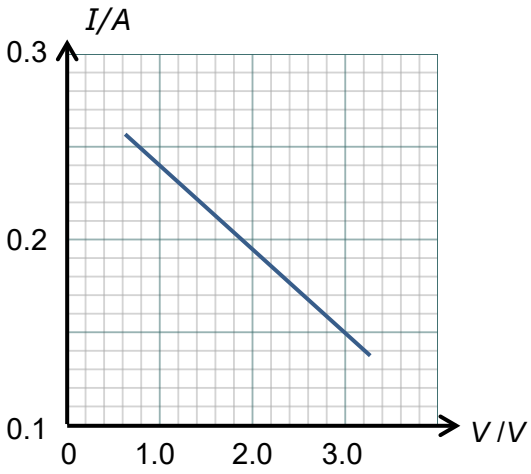


Fig. 9.3

Determine the resistance of the thermistor at 27 °C.

resistance = ... Ω [2]

(iii) Describe and explain what will happen to the potential difference across resistors A and B in the circuit when the temperature of the thermistor increases.

.....

.....

.....

.....[2]

(b) (i) Two identical lamps are connected to a d.c. and a.c. supply as shown in Fig. 9.4a and Fig. 9.4b respectively.

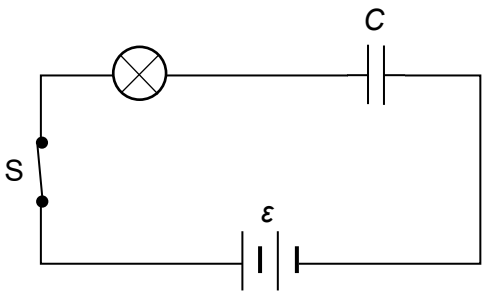


Fig 9.4a

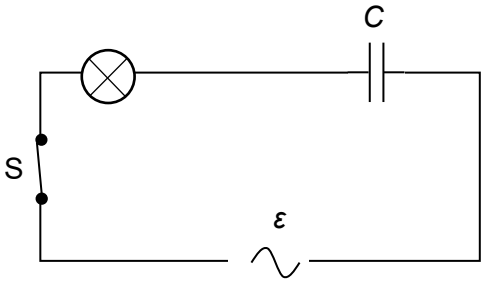


Fig 9.4b

State and explain which lamp will be turned on for a longer period of time. Assume that the time constant is large.

.....

.....

.....

.....

.....[3]

(ii) The charged capacitor of capacitance *C* is being discharged through a resistor of resistance *R*, as shown in Fig. 9.5. At time = 0 s, the capacitor is fully charged.

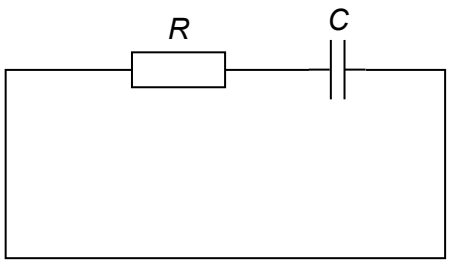


Fig. 9.5

The energy stored in the capacitor decreases with time as the capacitor discharges. Determine the time t, in terms of R and C, it takes for this stored energy to be discharged by 75% of its initial value.

t = ... [3]

(c) After a long time, the capacitor in Fig. 9.5 is fully discharged and a battery of e.m.f E of negligible internal resistance is used to recharge the capacitor in a circuit as shown in Fig. 9.6.

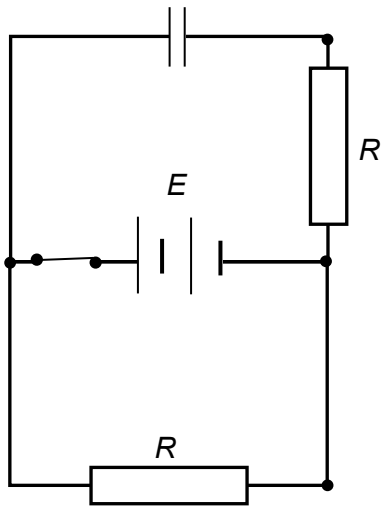


Fig. 9.6

(i) Determine the current, in terms of E and R, in the battery immediately after the switch is closed.

current = ... [2]

(ii) Determine the current, in terms of E and R, in the battery a long time after the switch is closed.

current = ... [2]

(iii) Hence suggest an expression for the time constant, in terms of R and C.

Explain your reasoning.

.....

.....

.....

.....

.....

.....[3]

[Total: 20]

END OF PAPER

5 A coil is placed in a region of magnetic field which is directed into the plane of the paper as shown in Fig. 5.1.

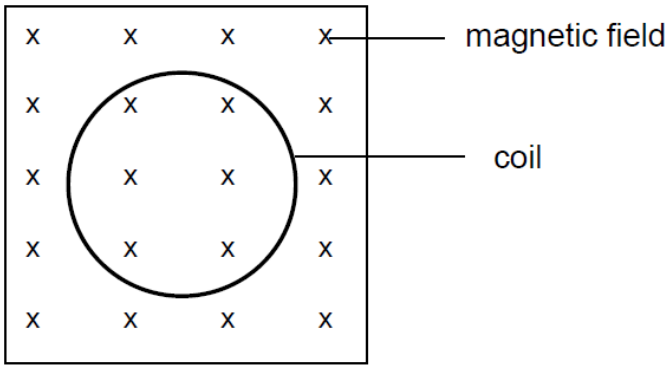


Fig. 5.1

(a) Define *magnetic flux*.

.....

.....

.....[1]

(b) **(i)** The magnitude of the flux density of the magnetic field is decreasing with respect to time. Explain, using Faraday's law, why there is an induced *current* in the coil.

.....

.....

.....[2]

(ii) Draw, on Fig. 5.1, the direction of the induced current. Explain, using Lenz's law, why the induced current is in this direction.

.....

.....

.....[2]

(c) A magnet of length *x* and magnetic flux density *B* is dropped vertically from rest with the north pole facing downwards. It is dropped from a reference line which is at a vertical height *h* above the top of a *long* solenoid of length *L* and uniform cross-sectional area as shown in Fig. 5.2. The length of the magnet *x* is much smaller than the length of the solenoid *L*.

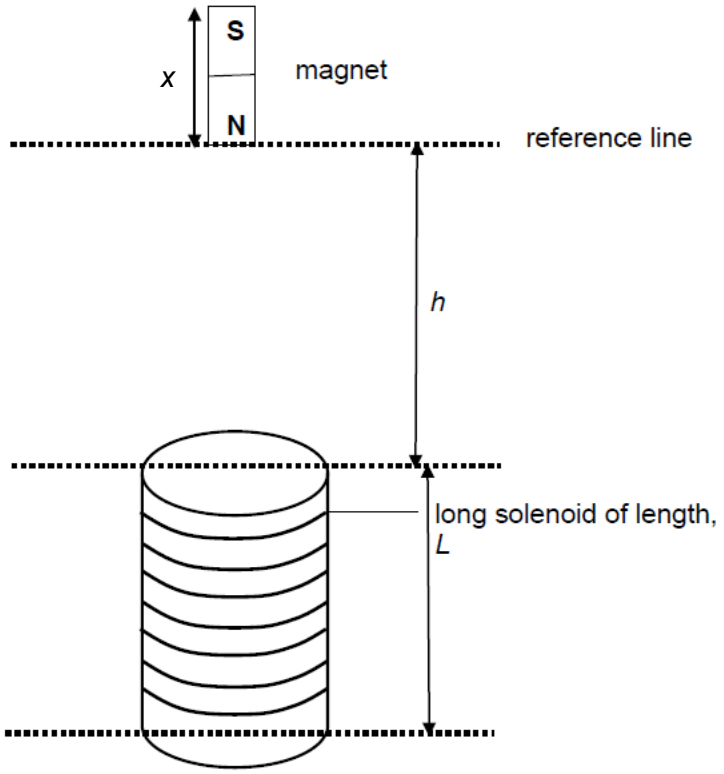


Fig. 5.2

(i) At time t = 0.0 s, the magnet is released from the reference line and allowed to pass through the entire length of the solenoid. Sketch, on Fig. 5.3, a graph to show the variation of the induced e.m.f E in the solenoid with time t. Label this graph as A. (Numerical values are not required and ignore the effects of air resistance.)



Fig. 5.3

[2]

(ii) State how your graph in **(c)(i)** would change if the same magnet is released with the south pole facing downwards instead.

.....

.....

.....[1]

(iii) The experiment is now repeated with the north pole of the magnet facing downwards, but released from a greater height above the solenoid. Sketch, on Fig. 5.3, a new graph to show the variation of the induced emf E in the solenoid with time t. Label this graph as B.

[2]

[Total: 10]

4 **(a)** Define *electric field strength*.

.....

.....

.....[1]

(b) An oil drop of mass 1.0×10^{-15} kg and carrying a charge of 1.6×10^{-19} C is placed in the middle of two horizontal parallel plates.

The separation between the plates is 40 mm and the the potential difference between the plates is 8.8 kV. The oil drop is seen to be attracted towards the upper plate because of its electric charge.

(i) Calculate the electric force experience by the oil drop.

force = ... N [2]

(ii) Initially the oil drop is 20 mm from either plate and is at rest. Ignoring air resistance, use energy considerations to calculate its speed when it reaches the upper plate.

speed = ... m s^{-1} [2]

[Total: 5]

5 During cosmic-ray experiments in the 1930s, researchers observed the track of a previously unknown positively charged particle in a cloud chamber. The particle was measured to have a charge of +1 *e* and is known as the muon.

The particle enters a uniform magnetic field of flux density 0.250 T with a speed of 2.40×10^{7} m s^{-1}. Its velocity makes an angle of 60° with the direction of the magnetic field, causing it to follow a helical path as seen in Fig. 5.1 below.

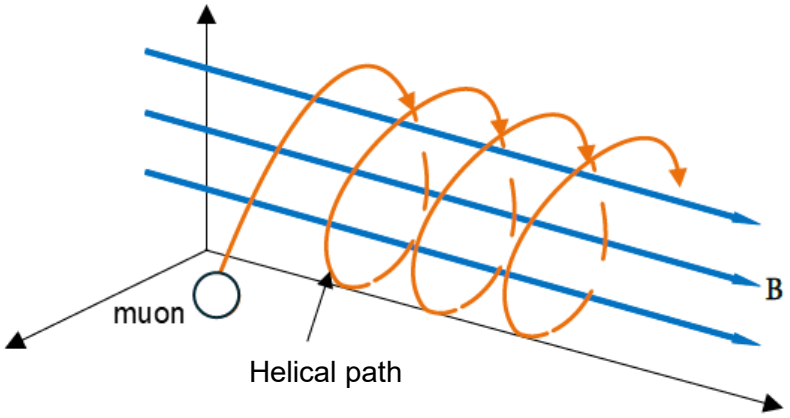


Fig. 5.1

Measurements of the track show that the muon took 2.96×10^{-8} s to complete one revolution.

(a) (i) Show that the radius of the helical path is 0.0979 m.

[2]

(ii) Determine the mass, m_{μ} of the muon.

m_{μ} = ... kg [2]

(b) Determine the distance travelled by the particle **parallel** to the magnetic field during one complete revolution.

distance = ... m [2]

(c) Before taking measurements, the researchers compensate for the vertical component of the Earth's magnetic field.

A short magnetic needle is suspended from a string 5.0 cm from a horizontal long straight wire, as shown in the top view in Fig. 5.2.

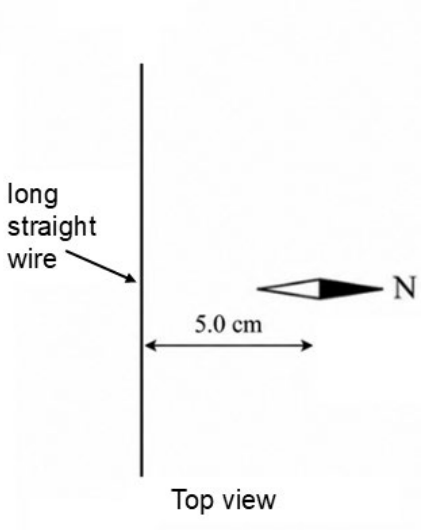


Fig. 5.2

When there is no current in the wire, the north pole of the needle dips at an angle of 6.0° below the horizontal, as shown in the side view in Fig. 5.3.

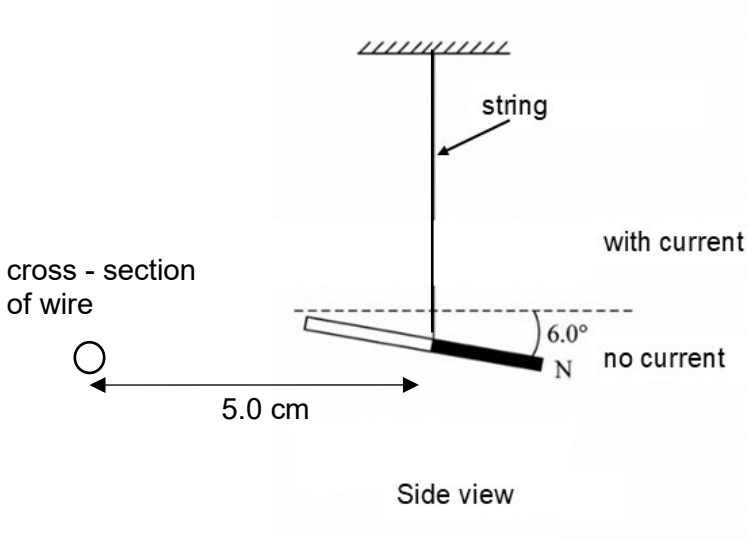


Fig. 5.3

A current of 0.47 A is then passed through the wire. The magnetic field produced by the current causes the needle to become horizontal, as shown in the side view in Fig. 5.4.

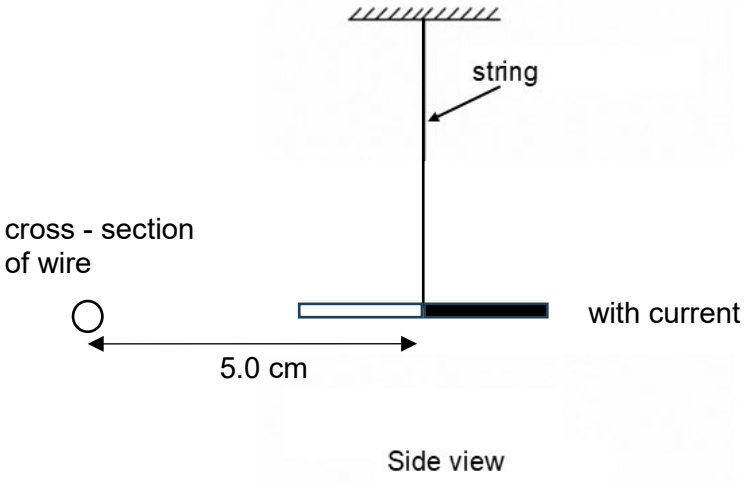


Fig. 5.4

(i) Show that the magnetic flux density produced by the wire, when the current is flowing through it, at the position of the short magnetic needle is 1.9×10^{-6} T.

[1]

(ii) Determine the magnitude of the Earth's magnetic flux density in the laboratory.

magnetic flux density = ...T [2]

Total: [9]

9 **(a)** Distinguish between *electromotive force e.m.f.* and *potential difference p.d.* using energy considerations.

.....

.....

.....[1]

(b) Fig. 9.1 shows a variable resistor and a fixed resistor R_1 connected to a battery with e.m.f E and internal resistance r. The variable resistor takes a range of values between 0 Ω to R_{max}.

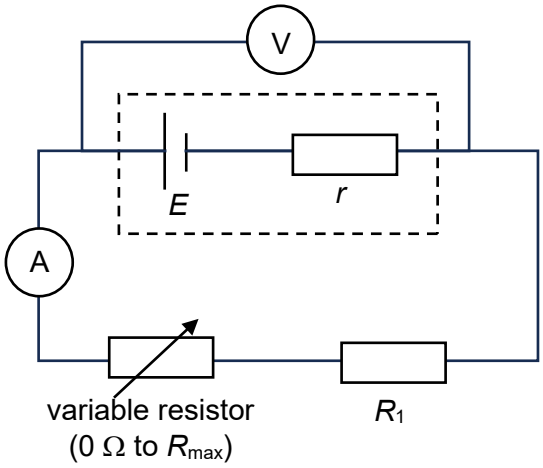


Fig. 9.1

The variation of the voltmeter reading V with current I through the ammeter as the variable resistor is varied over its **full range** of values is shown in Fig. 9.2.

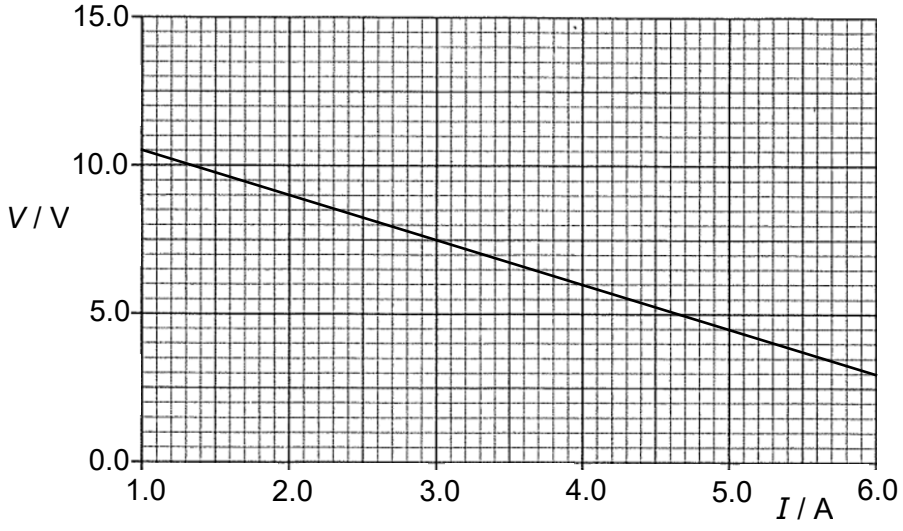


Fig. 9.2

(i) Using the graph, show that that internal resistance r is 1.5 Ω.

[3]

(ii) Calculate the e.m.f. E of the battery.

e.m.f. E = ... V [1]

(iii) Using Fig. 9.2, calculate the maximum resistance of the variable resistor R_{max}. Explain your working.

resistance R_{max} = ... Ω [3]

(c) The variable resistor is now set to 6.0 Ω, and a thermistor is connected in parallel with the variable resistor and R_1. The ammeter is also now connected in series with the thermistor as shown in Fig. 9.3.

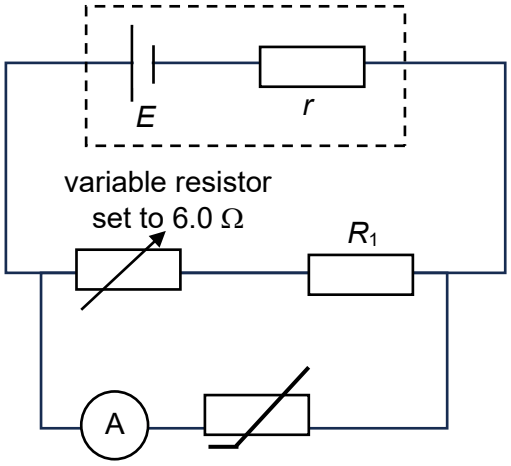


Fig. 9.3

(i) Explain how the potential difference across the thermistor changes when the temperature rises.

.....

.....

.....

.....[2]

(ii) Hence, or otherwise, determine how the current across the thermistor changes when the temperature rises.

.....

.....

.....

.....

.....

.....[3]

(d) Fig. 9.4 shows an alternating input voltage V_{IN} connected to a capacitor C and a resistor R.

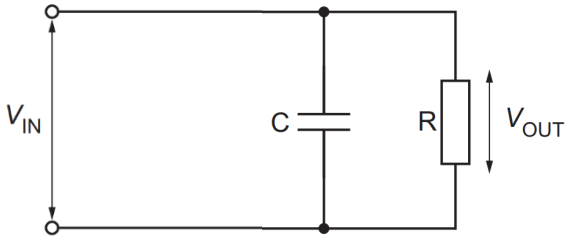


Fig. 9.4

The input voltage V_{IN} is a square wave. Fig 9.5 shows the variation of V_{IN} with time t.

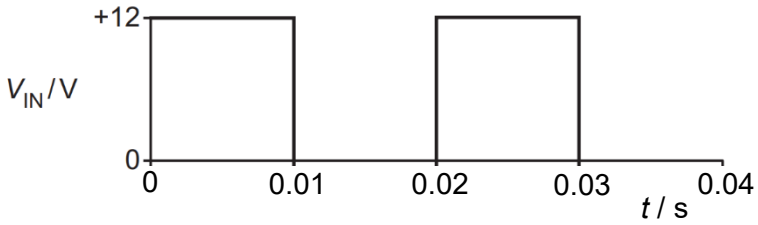


Fig. 9.5

Fig. 9.6 shows the variation of V_{OUT} with time t.

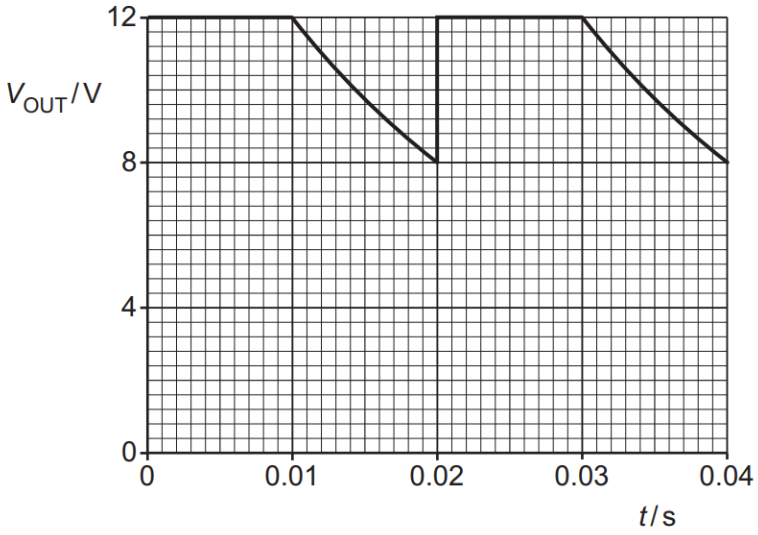


Fig. 9.6

The maximum energy stored in the capacitor is 0.041 J.

(i) Show that the capacitance of C is 570 μF.

[2]

(ii) Determine the resistance of R.

resistance = ... Ω [2]

(iii) Given two capacitors rated 800 μF and a third capacitor of 170 μF, suggest how capacitor *C* of 570 μF can be constructed from these three capacitors. Include a labelled diagram and show your workings clearly.

[3]

[Total: 20]